Faculty of Engineering, Sciences & Technology
BS (Software Engineering)

FYDP-1 Report

For

“SmartPool - Smart Carpooling System”

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DECLARATION

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Chapter – 1

1.0 Introduction:

Urban commuting in major Pakistani cities such as Karachi has become increasingly difficult due to a combination of rising fuel prices, recurring fuel-supply shortages, severe traffic congestion, and a general lack of secure and reliable shared-transport options. While ride-hailing services exist, they primarily operate as single-passenger, commercial taxi-style services rather than genuine cost-sharing carpooling platforms. Existing alternatives lack real-time route flexibility, fail to bridge the “trust gap” between strangers who must share a vehicle, and offer no meaningful incentive for environmentally responsible travel behaviour.

With advances in Artificial Intelligence, geo-spatial computing, and mobile technology, there is a strong opportunity to address these problems through an intelligent, secure, and incentive-driven carpooling platform. This project proposes SmartPool, an AI-Powered Smart Carpooling System that dynamically matches drivers and passengers along shared routes, verifies user identity to close the trust gap, calculates dynamic and fair fares, and rewards sustainable travel through a gamified carbon-credit system.

A. Project Summary:

SmartPool is an AI-powered, mobile-first carpooling platform designed for urban commuters in Karachi to mitigate rising fuel costs, severe traffic congestion, and the "trust gap" inherent in traditional shared transport. Moving away from conventional commercial ride-hailing structures, the system focuses on a fair, non-profit cost-distribution model. It automatically calculates total trip costs based on vehicle fuel economy, real-time petrol prices, and maintenance variables, splitting expenses equally among all vehicle occupants.

The technical architecture combines Intelligent Systems, Software Engineering, and Human-Computer Interaction. SmartPool utilizes Genetic Algorithms to handle dynamic route matching and pinpoint "Virtual Meeting Points," minimizing driver detours while maximizing passenger convenience. Developed using an Agile Scrum methodology, the platform integrates a Flutter/React frontend with a Node.js backend, leveraging PostgreSQL/PostGIS for robust geo-spatial querying.

To ensure user safety and prevent data fraud without requiring restrictive institutional emails, the platform implements a tiered security pipeline:

- **Biometric Identity Integrity:** Uses Google ML Kit for on-device facial geometry matching, cross-referencing live selfies against extracted CNIC photos.
- **Anti-Spoofing Operations:** Employs real-time map snapping to bind GPS coordinates to road networks and utilizes dual-device proximity validation to eliminate mileage fraud.
- **Secure Financial Handshakes:** Features an automated escrow system integrated with local mobile wallets (Easypaisa/JazzCash) alongside unalterable, read-only ride records.

B. Objective, Benefits and Deliverables

1- Objectives

- **Optimize Urban Mobility:** Implement Genetic Algorithms to dynamically match riders and drivers, establishing optimized "Virtual Meeting Points" to minimize detours.
- **Bridge the Trust Gap:** Establish a secure, universal onboarding pipeline using CNIC OCR, on-device biometric face matching, and SIM-based OTP verification.
- **Eliminate Fraud:** Enforce data integrity through real-time GPS map snapping and dual-device proximity validation to block mileage and identity fraud.
- **Facilitate Fair Cost-Sharing:** Deploy an automated pricing engine to execute proportional, non-profit seat-splitting based on real-time fuel and maintenance costs.
- **Promote Green Travel:** Encourage sustainable commuting habits via a Python-modeled, gamified carbon-credit ledger.

2- Benefits

- **Cost Efficiency:** Reduces individual commuting expenses for citizens in Karachi by splitting exact trip costs directly among vehicle occupants.
- **Enhanced Safety and Trust:** Ensures accountability by validating every user against a real identity (CNIC/Selfie) without relying on institutional emails.
- **Reduced Congestion:** Displaces single-occupancy vehicles from road networks.
- **Environmental Impact:** Lowers urban carbon emissions and rewards eco-friendly choices through active sustainability tracking .
- **Dispute Prevention:** Eradicates fare tampering and financial arguments via automated wallet escrow (Easypaisa/JazzCash) and immutable ride receipts.

3- Deliverables

- **Cross-Platform Mobile App:** A functional Flutter/React Native application featuring integrated AI matching and live geospatial tracking.
- **Secure Database Infrastructure:** A PostgreSQL/PostGIS backend configured with read-only constraints for unalterable financial and ride records.
- **Verification & Anti-Spoofing Pipeline:** Integrated Google ML Kit modules for facial matching and dual-device location validation protocols.
- **Technical Documentation:** Comprehensive Software Development Life Cycle (SDLC) documentation, system architecture designs, and final FYP reporting.

C. Plan:

Development / Research Methodology:

The project will follow an Iterative Agile (Scrum) Methodology to allow for continuous feedback and refinement of AI models:

1. Requirements & Analysis: Identifying local commute patterns in Karachi and defining the constraints for AI matching and security protocols.

2. System Design: Architecting the Flutter frontend, Node.js backend, and the integration of PostGIS for geo-spatial querying.

3. Implementation (Sprints):

- **Sprint 1**: Core Authentication & User Profile setup.
- **Sprint 4:** Integration of the AI Route Optimization engine.
- **Sprint 3**: Building the "Green" Wallet and Payment gateway.

4. Testing: Performing unit, integration, and user-acceptance testing to ensure AI accuracy and financial security.

5. Deployment & Documentation: Final release of the Android app and completion of the technical documentation for FYP submission.

Resources Involved:

Standard Development: VS Code on Kubuntu, Flutter SDK/ReactNative, Node.js, and GitHub for version control.

Route Optimization: Google Maps Platform (Directions & Distance Matrix APIs) to provide real-time data for the Genetic Algorithm.

Trust & Security: Google ML Kit for on-device face detection and Natural Language Toolkit (NLTK/TextBlob) for sentiment analysis of reviews.

Sustainability Tracking: Python-based Models hosted on FastAPI to calculate carbon offsets based on vehicle engine specifications.

Database & Payment: PostgreSQL/PostGIS for geographical queries and Easypaisa/JazzCash SDKs for secure transactions.

SmartPool User Flow Architecture

1. Universal Onboarding & Verification Pipeline

- **App Installation:** The user downloads the Flutter/React Native application.
- **Phone Verification:** The user inputs their mobile number and completes a SIM-based OTP verification.
- **Identity Extraction:** The user takes a photo of their CNIC; the system runs OCR to automatically extract the legal name and CNIC number.
- **Biometric Match:** The user takes a live selfie. On-device Google ML Kit face detection analyzes facial geometry and matches it against the extracted CNIC photo to verify identity integrity.
- **Account Activation:** Upon a successful match, a secure, verified profile is created within the PostgreSQL database.

2. Profile Setup (Driver-Specific)

- **Vehicle Registration:** Verified users choosing to drive must input vehicle details, including engine specifications and vehicle fuel economy (FE).
- **Wallet Linkage:** The driver links their local digital wallet (Easypaisa/JazzCash) to receive automated cost-share payouts.

3. Ride Scheduling & AI Optimization Engine

- **Ride Initiation:**
  - **Driver:** Posts a planned route with an initial departure time and destination.
  - **Passenger:** Requests a ride by inputting their current location and desired drop-off destination.
- **Geospatial Processing:** PostGIS queries index the coordinates to locate active, overlapping routes in Karachi.
- **Dynamic Route Matching:** The backend processes data through Google Maps Directions & Distance Matrix APIs. A Genetic Algorithm optimization engine computes minimal detours for the driver while identifying exact **Virtual Meeting Points** for the passengers.

Fare Calculation & Lock: The dynamic pricing engine computes the total trip cost (CT) and locks the segment-based fare splitting:

$$\text{Passenger Share} = \frac{C_{\text{Segment}}}{\text{Occupants in Segment}}$$

Match Confirmation: Passengers accept the fare lock, and money is temporarily moved from their mobile wallet into the platform's secure **automated escrow system**.

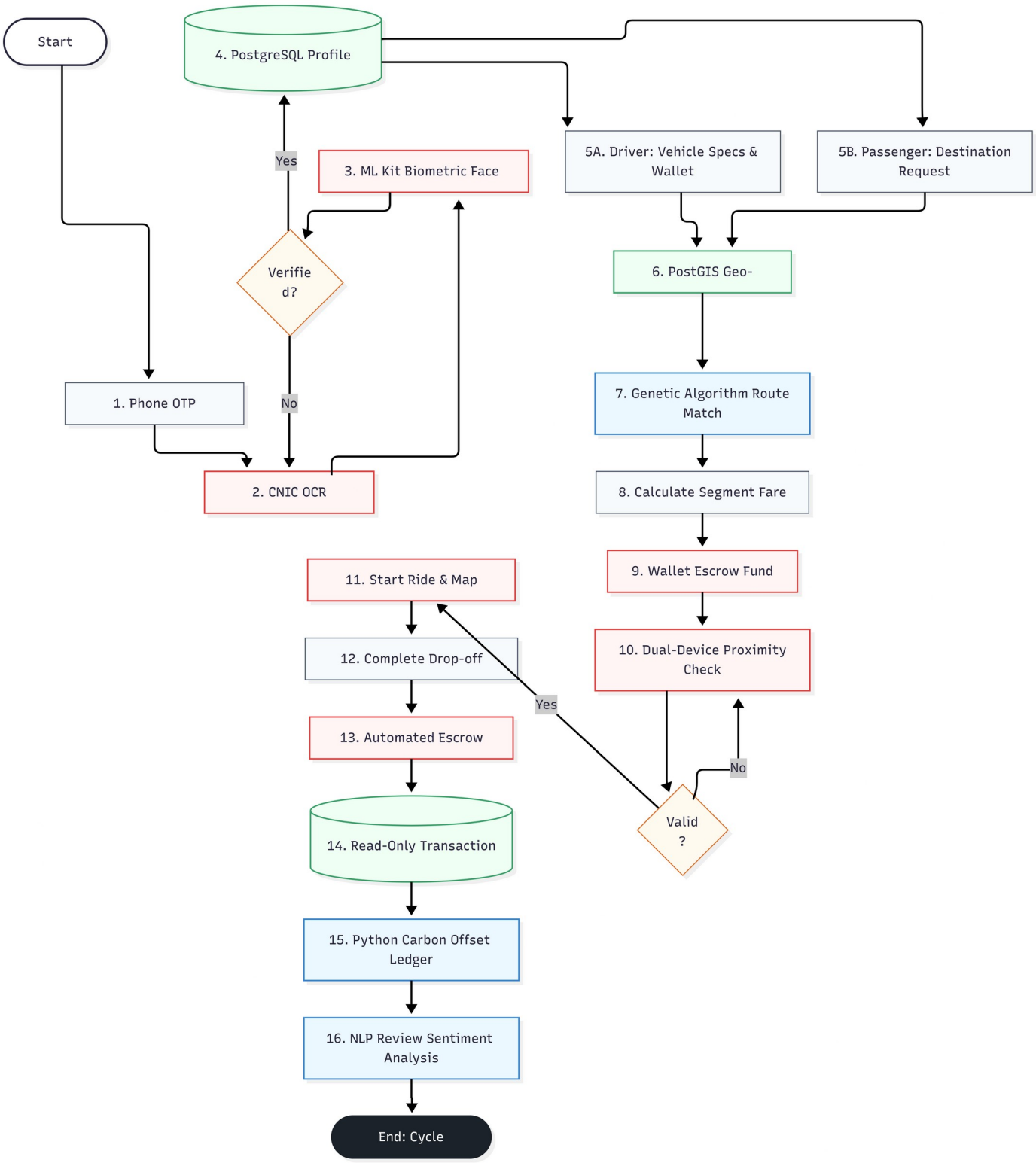
4. Live Commute & Anti-Spoofing Operations

- **Transit to Virtual Meeting Point:** Both parties navigate to the designated Virtual Meeting Point using the in-app map interface.
- **Proximity Validation:** When the vehicle arrives, **Dual-Device Proximity Validation** cross-references the driver's and passenger's live GPS coordinates via Bluetooth/location APIs to verify physical presence.
- **Ride Commencement:** The driver starts the trip.
- **Route Invariant Tracking:** During transit, **Map Snapping** continuously binds live coordinates to real Karachi road networks, ignoring GPS glitches and preventing fraudulent mileage inflation.

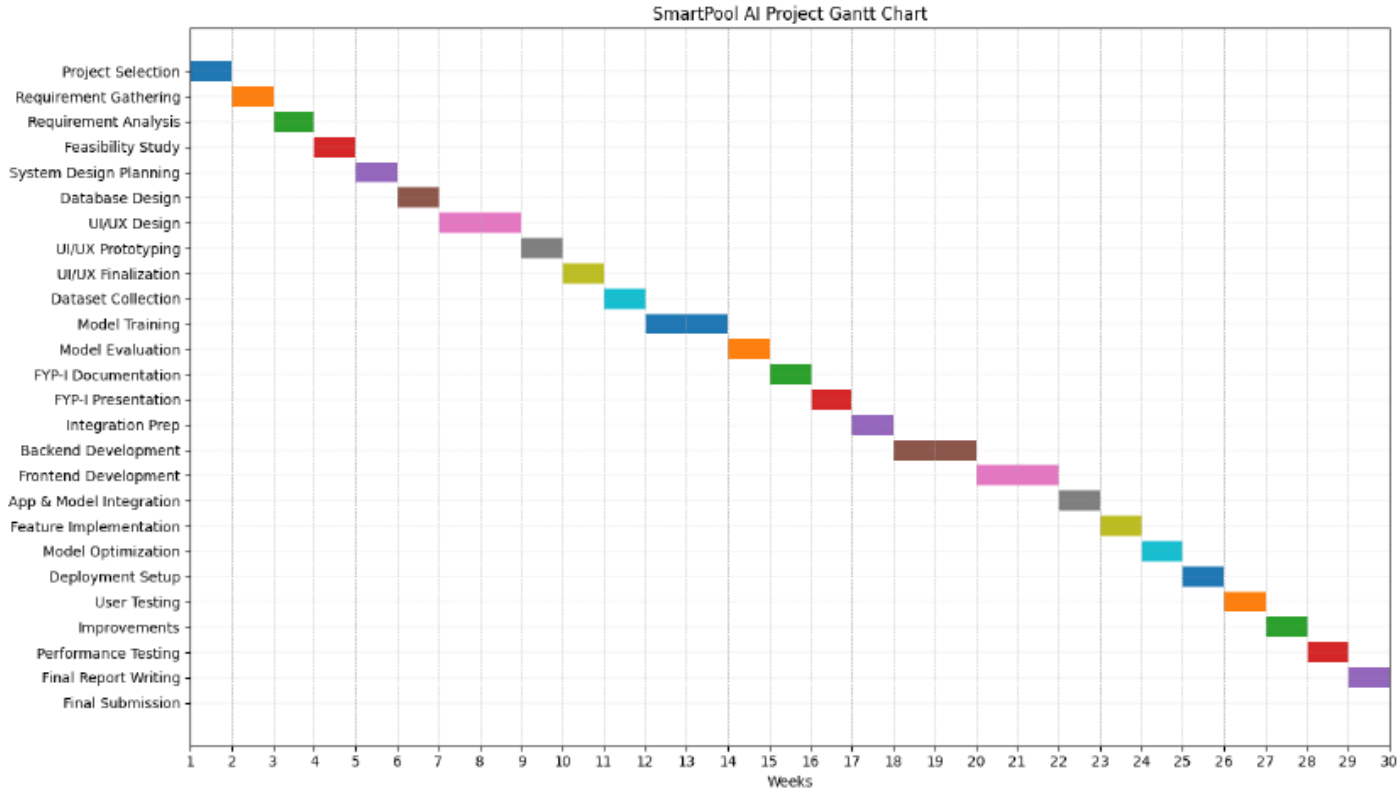
5. Drop-off, Escrow Release & Sustainability Ledger

- **Destination Reached:** The passenger arrives at their drop-off segment. The driver marks the segment complete.
- **Escrow Settlement:** The system confirms successful tracking data and automatically releases the exact calculated seat-share from the escrow holding directly into the driver's connected wallet.
- **Read-Only Ledgering:** The finalized receipt is written to a read-only PostgreSQL table constraint, guaranteeing unalterable transaction history.
- **Carbon-Credit Ledger:** A Python-based sustainability microservice parses the vehicle's profile and actual distance driven, updating the passenger's gamified wallet with earned green carbon credits.
- **Trust & Safety Feedback:** Both users review each other. The backend passes text through an NLP engine (NLTK/TextBlob) to perform sentiment analysis, updating each user's behavioral trust-score.

Flowchart


Start
4. PostgreSQL Profile
Yes
3. ML Kit Biometric Face
Verifie d?
No
1. Phone OTP
2. CNIC OCR
5A. Driver: Vehicle Specs & Wallet
5B. Passenger: Destination Request
6. PostGIS Geo-
7. Genetic Algorithm Route Match
8. Calculate Segment Fare
9. Wallet Escrow Fund
10. Dual-Device Proximity Check
Valid ?
No
Yes
11. Start Ride & Map
12. Complete Drop-off
13. Automated Escrow
14. Read-Only Transaction
15. Python Carbon Offset Ledger
16. NLP Review Sentiment Analysis
End: Cycle

Gantt Chart


SmartPool AI Project Gantt Chart
Project Selection
Requirement Gathering
Requirement Analysis
Feasibility Study
System Design Planning
Database Design
UI/UX Design
UI/UX Prototyping
UI/UX Finalization
Dataset Collection
Model Training
Model Evaluation
FYP-I Documentation
FYP-I Presentation
Integration Prep
Backend Development
Frontend Development
App & Model Integration
Feature Implementation
Model Optimization
Deployment Setup
User Testing
Improvements
Performance Testing
Final Report Writing
Final Submission
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
Weeks

A. Key Milestones of the Project / Project Management:					
S. No.	**Elapsed time since start of the project**	**Start Date**	**End Date**	**Milestone**	**Deliverable**
1.	Week 01 – 04	May 7, 2026	June 3, 2026	Define project scope, secure supervisor approval, and complete requirement analysis by identifying Karachi urban commute constraints, AI route matching parameters, biometric verification needs, and local SDK technical viability.	Approved FYP Proposal Form (FYDP-Form1), comprehensive Feasibility Study Report, and finalized Software Requirements Specification (SRS) document.
2.	Week 05	June 4, 2026	June 10, 2026	Map out high-level software system design, components, and communication protocols.	High-Level Architecture Diagrams.
3.	Week 06	June 11, 2026	June 17, 2026	Plan PostgreSQL/PostGIS database structure, attributes, and relational foreign keys.	Database Entity-Relationship Diagrams (ERD).
4.	Week 07	June 18, 2026	June 24, 2026	Create wireframes focusing on the geospatial map layout and universal onboarding screens.	Low-Fidelity Wireframes.
5.	Week 08	June 25, 2026	July 1, 2026	Transform wireframes into high-fidelity mockups mapping user-flows for drivers and riders.	High-Fidelity UI/UX Mockups.
6.	Week 09	July 2, 2026	July 8, 2026	Construct click-through front-end screens to simulate live navigation transitions.	Interactive UI/UX Prototype.
7.	Week 10	July 9, 2026	July 15, 2026	Present UI/UX layouts to the supervisor and finalize front-end structural components.	Approved and Signed-off UI/UX Designs.

S. No.	Elapsed time since start of the project	Start Date	End Date	Milestone	Deliverable
8.	Week 11	July 16, 2026	July 22, 2026	Collect, scrub, and catalog local coordinate and road network data for Karachi.	Cleaned Geospatial Testing Dataset.
9.	Week 12	July 23, 2026	July 29, 2026	Set up Python/FastAPI environment and select baseline parameters for the Genetic Algorithm.	Algorithmic Testing Environment.
10.	Week 13	July 30, 2026	Aug 5, 2026	Run Phase 1 training on the route optimization engine to isolate detour mitigation vectors.	Initial Deployed Route Optimization Model.
11.	Week 14	Aug 6, 2026	Aug 12, 2026	Audit algorithm performance against real-world test paths to track optimization metrics.	AI Model Performance Evaluation Report.
12.	Week 15	Aug 13, 2026	Aug 19, 2026	Compile all theoretical research, design diagrams, and modeling metrics into documentation.	Complete FYP-I Draft Thesis Report.
13.	Week 16	Aug 20, 2026	Aug 26, 2026	Assemble defense assets and execute formal academic presentation before the panel.	Final FYP-I Slides and Evaluated Project Log.
14.	Week 17	Aug 27, 2026	Sept 2, 2026	Initialize the codebase, establishing Node.js endpoints and mapping out pipeline connections.	Technical Architectural Integration Plan.
15.	Week 18	Sept 3, 2026	Sept 9, 2026	Program core Node.js server engines for profile and operational ride request logic.	Functional Backend Server Modules.
16.	Week 19	Sept 10, 2026	Sept 16, 2026	Spin up relational tables in PostgreSQL, enabling spatial query indices using PostGIS extensions.	Configured Production Geospatial Database.
17.	Week 20	Sept 17, 2026	Sept 23, 2026	Code primary functional interfaces in Flutter/React Native, starting with onboarding views.	Executable Front-end Interface Layouts.

S. No.	Elapsed time since start of the project	Start Date	End Date	Milestone	Deliverable
18.	Week 21	Sept 24, 2026	Sept 30, 2026	Code dynamic map interfaces using Google Maps Directions & Distance Matrix APIs.	Live-Map Interactive Client Engine.
19.	Week 22	Oct 1, 2026	Oct 7, 2026	Connect Flutter client requests to Node.js backend pipelines, validating API data loops.	Integration Phase Testing Documentation.
20.	Week 23	Oct 8, 2026	Oct 14, 2026	Deploy local SDK wallet integrations and verify on-device Google ML Kit face recognition modules.	Verified Universal Security Pipeline.
21.	Week 24	Oct 15, 2026	Oct 21, 2026	Write real-time GPS Map Snapping algorithms and dual-device proximity tracking layers.	Anti-Spoofing Operations Subsystem.
22.	Week 25	Oct 22, 2026	Oct 28, 2026	Host the system on live sandbox servers, establishing staging environments.	Live Staging Deployment Build.
23.	Week 26	Oct 29, 2026	Nov 4, 2026	Distribute early client applications to closed beta testers across local Karachi trial routes.	Field Alpha-User Evaluation Report.
24.	Week 27	Nov 5, 2026	Nov 11, 2026	Remediate bugs discovered in testing loops, optimize API queries, and seal UI gaps.	Patched Stable Android App Release Binary (APK).
25.	Week 28	Nov 12, 2026	Nov 18, 2026	Measure processing latencies, AI throughput, and database transaction performance logs.	Final Software Performance Evaluation Audit.
26.	Week 29	Nov 19, 2026	Nov 25, 2026	Compose final thesis content spanning source code validation, system tests, and architecture.	Comprehensive Final Thesis Manuscript Draft.

S. No.	Elapsed time since start of the project	Start Date	End Date	Milestone	Deliverable
27.	Week 30	Nov 26, 2026	Feb 15, 2027	Review documentation, polish delivery assets, and hold project defense prep *(includes built-in evaluation buffer)*.	**Final Submission Package:** Deployed platform, manual, demo videos, and slides.
J. Final deliverable of the project					
☐ Hardware system		☐	☐	☐ Software system	☐ HW/SW integrated system
☐ Software simulation results		☐	☐	☐ Comparative study	✅ Mobile App
☐ Web App		☐	☐	☐ Other (please specify)	

J. Equipment required for the project with Estimated Cost:

Category	Item / Service	Purpose	Cost (PKR)
Hardware	Development Laptop	Host software stack, compile Flutter builds, and run PostGIS DB.	Rs. 165,000
Hardware	Android Test Phone	Physical testing of GPS tracking, ML Kit biometric match, and OTPs.	Rs. 35,000
Cloud Infra	VPS Server Hosting	12-month hosting for Node.js APIs and Python AI microservices.	Rs. 24,000
Cloud Infra	Managed Database	12-month hosting for production PostgreSQL/PostGIS database instance.	Rs. 18,000
APIs	Google Maps Platform	Production quota for Directions and Distance Matrix optimization APIs.	Rs. 15,000
APIs	SMS Gateway	Commercial API volume to dispatch secure registration OTP tokens.	Rs. 10,000
Security	SSL/TLS Certificate	Production-grade encryption for API traffic and wallet transaction data.	Rs. 5,000
TOTAL		**Comprehensive Project Production Budget**	**Rs. 272,000**

B. How many GAs are you covering in this Proposal? List all GAs

GA-2 Knowledge for Solving Computing Problems
GA-3 Problem Analysis
GA-4 Design/Development of Solutions
GA-5 Modern Tool Usage
GA-8 Computing Professionalism and Society
GA-9 Ethics
GA-6 Individual and Teamwork
GA-7 Communication
GA-10 Life-long Learning
GA-1 Academic Education

C. How many SDGs are you covering in this Proposal? List all SDGs	
1.No poverty	**10. Reduced inequalities**
2.Zero hunger	**11. Sustainable cities and communities**
3.Good health and well being	**12. Responsible consumption and production**
4.Quality education	**13. Climate action**
5.Gender equality	**14. Life below water**
6.Clean water and sanitation	**15. Life on land**
7.Affordable and clean energy	**16. Peace, justice and strong institutions**
8.Decent work and economic growth	**17. Partnership for the goals.**
9.Industry innovation and infrastructure	

Wps through this project:

WP's	**How it is addressed in this Project**
WP1: Depth of Knowledge Required	The project spans four distinct ACM computing domains: Intelligent Systems (IS) for optimization, Software Engineering (SE) for full-lifecycle development, Information Management (IM) for geospatial data indexing, and Information Assurance & Security (IAS) for identity verification.
WP2: Range of Conflicting Requirements	The platform is forced to run computationally heavy, real-time algorithms (Genetic Algorithms and continuous Map Snapping) alongside active on-device biometric processing. It explicitly balances these high-performance demands against strict mobile constraints, such as low battery consumption, optimized mobile data usage, and user privacy.

WP's	**How it is addressed in this Project**

WP3: Depth of Analysis Required	There is no standard out-of-the-box solution for dynamic multi-passenger ride matching. The project solves this non-trivial, multi-objective optimization problem by designing custom Genetic Algorithms to maximize vehicle capacity and passenger convenience while minimizing driver detour paths.
WP4: Familiarity of Issues	The software moves past basic database storage by implementing localized, on-device face detection models via TensorFlow Lite (Google ML Kit) for cross-verifying live selfies against official CNIC data. It also introduces a localized custom carbon-credit ledger integrated into the local digital payment ecosystem.
WP5: Extent of Applicable Codes	Development requires navigating outside standard web frameworks to comply with local transportation regulations, data protection practices for handling legal identities (CNICs), and securing financial API integrations with local commercial wallet providers (Easypaisa and JazzCash).
WP6: Extent of Stakeholder Involvement and Conflicting Requirements	The platform synchronizes diverse stakeholder groups with competing needs. Drivers require compensation for maintenance and exact fuel costs without making a commercial profit, passengers demand low fares and zero delays, university administrations look for academic/institutional validation, and payment gateways mandate zero-trust fraud safety.
WP7: Interdependence	The application is highly interdependent. If the geospatial map snapping or proximity tracking layers fail, the dynamic segment-fare calculation engine receives corrupted inputs. This breaks the automated mobile wallet escrow payouts, which in turn invalidates the Python-based sustainability ledger. Every single sub-microservice relies on the absolute data integrity of the other.

Chapter # 02

Related System Analysis

1- BlaBlaCar (France)

Introduction:

BlaBlaCar is a French long-distance carpooling platform founded in 2006, currently operating in over 20 countries with more than 100 million members. It connects private drivers who have empty seats on planned intercity trips with passengers travelling in the same direction, allowing costs to be shared between them. It is primarily a web and mobile-based platform focused on intercity travel rather than daily urban commuting.

Working Model:

BlaBlaCar operates on a publish-and-search model. Drivers post planned trips with their route, departure time, available seats, and a suggested price. Passengers search for matching trips and send a booking request. Once the driver accepts, payment is processed through the platform. Identity verification is done through email, phone number, and optionally a government-issued ID. The platform uses a star-based rating system where both drivers and passengers review each other after every trip to build trust over time.

Advantages:

- Large, established user base provides wide route coverage across countries.
- Integrated payment system with clear cost-splitting between driver and passengers.
- Rating and review system creates a layer of social accountability.
- Available as both web and mobile app, making it accessible on multiple platforms.

Disadvantages:

- Primarily designed for long-distance intercity travel; does not support daily urban commuting routes.
- No real-time AI-based dynamic route matching — drivers post fixed trips and passengers must conform to them.
- Identity verification is not stringent for local markets; relies largely on social reputation rather than biometric or CNIC-based verification.
- No support for local payment methods such as Easypaisa or JazzCash, making it inaccessible to

the Pakistani general public.

- No GPS anti-fraud or route snapping mechanism to prevent mileage manipulation.
- No environmental incentive system such as carbon credit tracking.

Summary:

BlaBlaCar is a well-established carpooling solution that has proven the commercial viability of shared travel. However, it is built for intercity routes in developed markets and is not adapted for urban daily commuting in cities like Karachi. It lacks real-time AI matching, local payment integration, biometric identity verification, and any GPS-based fraud prevention — all of which are core pillars of SmartPool. The contribution of SmartPool over BlaBlaCar lies in its dynamic AI-driven route optimization for daily urban use, CNIC-based identity integrity, local wallet integration, and its gamified environmental incentive layer — none of which BlaBlaCar offers.

2- Careem (Dubai)

Introduction:

Careem is a Dubai-based ride-hailing platform founded in 2012, widely used across the Middle East, North Africa, and South Asia, including Pakistan. It operates as an on-demand transportation service connecting passengers with professional or semi-professional drivers through a mobile application. Careem was acquired by Uber in 2019 and has since expanded into a super-app offering food delivery, payments, and other services alongside its core ride-hailing product.

Working Model:

Careem operates on a real-time demand-supply model. A passenger opens the app, enters a destination, and the system automatically matches them with the nearest available driver using GPS. Fares are calculated dynamically based on distance, time, and surge pricing during peak hours. Payment is supported through cash, card, and Careem Credits (an in-app wallet). The platform tracks rides in real-time via GPS and provides an in-app SOS button for safety. Driver registration requires a vehicle document check and a basic background screening process.

Advantages:

- Real-time GPS tracking provides live ride visibility for both passengers and emergency contacts.
- Supports local payment methods including cash and digital wallets relevant to the Pakistani market.
- Established safety features including SOS buttons and trip sharing with trusted contacts.
- Large existing user base and strong brand trust across Karachi and other Pakistani cities.
- Surge pricing model responds dynamically to demand fluctuations.

Disadvantages:

- Careem is a commercial ride-hailing service, not a carpooling platform — the driver is a paid professional, not a fellow commuter sharing costs.
- No cost-splitting among multiple passengers travelling the same route; each booking is treated as a separate commercial transaction.
- Fares are significantly higher than shared transport due to driver profit margins and platform commissions, making it financially unsustainable for daily commuters.
- No AI-driven route optimization to dynamically group passengers heading in the same direction.
- No biometric identity verification such as CNIC or facial matching for general public onboarding.
- No environmental tracking or carbon credit incentives for sustainable travel choices.
- Drivers are incentivized to maximize trip count for earnings, not to reduce congestion or fuel consumption.

Summary: Careem addresses the urban mobility problem through a commercial model that prioritizes convenience for individual passengers rather than shared, cost-effective commuting. While it excels in real-time tracking and local market adaptability, it is fundamentally different from SmartPool in purpose and economics. SmartPool is not a commercial ride-hailing service — it is a peer-to-peer cost-sharing platform where a regular commuter driver recovers fuel and maintenance costs by sharing seats with passengers on the same route. This distinction means SmartPool can offer significantly lower per-trip costs, reduce the number of vehicles on the road, and promote environmentally conscious travel — outcomes that a profit-driven model like Careem is structurally unable to deliver.

3- Uber (USA)

Introduction:

Uber Pool, rebranded as Uber Share in several markets, is a shared ride feature introduced by Uber in 2014 as part of its broader ride-hailing platform. It was designed to reduce individual trip costs by algorithmically grouping multiple passengers travelling in similar directions into a single vehicle. It operates across major cities globally and represents one of the closest commercial attempts at combining ride-hailing convenience with carpooling economics. However, the feature has been suspended or scaled back in several markets, including Pakistan, due to driver resistance and operational complexity.

Working Model:

When a passenger selects Uber Pool, the system uses a matching algorithm to find other passengers with overlapping routes within a short time window. The algorithm calculates optimal pickup and drop-off sequencing to minimize total detour for the driver. Fares are calculated at a discounted rate compared to a standard UberX trip, with the discount reflecting the shared nature of the ride. The driver follows a dynamically generated multi-stop route shown in the app. Payment is fully automated through card or in-app wallet. All passengers and drivers go through Uber's standard onboarding process, which includes license verification for drivers and phone/email registration for passengers.

Advantages:

- Algorithmic passenger grouping reduces per-person fare compared to solo rides.
- Dynamic multi-stop route generation provides a working model for shared urban mobility.
- Fully automated fare calculation and payment removes manual negotiation between parties.
- Real-time GPS tracking and in-app communication provide a degree of safety and transparency.
- Proven at scale in multiple major cities, demonstrating technical and operational viability of shared rides.

Disadvantages:

- Like Careem, Uber Pool is still a commercial service — the driver is a paid professional, not a peer commuter, so costs remain higher than true carpooling.
- The matching algorithm prioritizes Uber's revenue efficiency over passenger convenience, often resulting in long detours and extended travel times.
- No biometric or government ID-based verification for passengers, creating a trust gap particularly relevant in markets like Pakistan.
- Matching is reactive and opportunistic — it only groups passengers who happen to request rides simultaneously, rather than proactively planning shared routes in advance.
- No support for local Pakistani payment methods such as Easypaisa or JazzCash in its current form.
- No carbon credit system or environmental incentive layer for users choosing to share rides.
- Feature availability is inconsistent across markets and has been pulled from Pakistan entirely.

Summary: Uber Pool is the most technically comparable existing system to SmartPool in terms of its shared routing logic and algorithmic passenger grouping. However, it remains anchored to the commercial ride-hailing model where drivers earn profit per trip rather than simply recovering travel costs. SmartPool's key distinction is that it operates in the peer-to-peer space — a regular commuter driving their own vehicle on their regular route recovers only fuel and maintenance costs while passengers benefit from drastically lower fares. Additionally, SmartPool improves upon Uber Pool's matching approach by using Genetic Algorithms to pre-plan Virtual Meeting Points, incorporates CNIC-based biometric identity verification for the Pakistani public, integrates local payment wallets, and adds a gamified carbon credit system — layers of functionality that Uber Pool does not offer and that are specifically designed for the Karachi commuter context.

4- DiDi (China)

Introduction:

DiDi Chuxing, commonly known as DiDi, is a Chinese transportation platform founded in 2012 and currently the dominant ride-hailing service in China with over 550 million users. Within its suite of services, DiDi offers a dedicated Carpool feature (顺风车, Shunfengche) that is specifically designed for peer-to-peer ride sharing rather than commercial ride-hailing. In this model, everyday drivers heading to a destination post their trip in advance, and passengers travelling the same way book a seat and share the actual travel cost. This makes DiDi Carpool structurally the closest international equivalent to SmartPool's core concept, making it a highly relevant system for comparison.

Working Model:

DiDi Carpool operates on a pre-planned trip posting model. A driver registers their regular commute route, departure time, and available seats ahead of time. The platform's algorithm then matches waiting passengers whose origin and destination fall along or close to that route. Fare is calculated based on actual distance and government-regulated cost-per-kilometer rates to ensure drivers do not profit commercially. Payment is processed automatically through DiDi's integrated wallet (linked to WeChat Pay and Alipay). The platform uses real-time GPS tracking throughout the trip. Following a serious safety incident in 2018, DiDi overhauled the feature with enhanced security measures including mandatory facial recognition before each trip, in-trip audio recording with passenger consent, an emergency SOS button, and a friends-and-family live tracking feature.

Advantages:

- Genuine peer-to-peer cost sharing model where drivers recover costs rather than earn profit, directly aligned with SmartPool's philosophy.
- Mandatory pre-trip facial recognition verification significantly reduces identity fraud and increases passenger safety.
- Government-regulated fare caps prevent driver overcharging and maintain the non-commercial nature of the service.
- Robust safety infrastructure including SOS, live tracking for trusted contacts, and in-trip recording.
- Massive scale demonstrates that AI-assisted peer carpooling is operationally viable in dense urban environments.
- Pre-planned trip model allows better route optimization compared to reactive on-demand matching.

Disadvantages:

- Operates exclusively within China's digital ecosystem, deeply integrated with WeChat Pay and Alipay, making it entirely inaccessible and non-transferable to the Pakistani market.
- The trust and safety overhaul came only after fatal safety incidents, indicating the original system had critical gaps that required a crisis to address — SmartPool addresses these proactively from inception.
- Matching is still largely route-conformity based, requiring passengers to fit the driver's pre-set route rather than dynamically computing optimal meeting points to minimize detour for all parties.
- No gamified environmental incentive system such as carbon credit rewards for users choosing to carpool.
- Relies on a highly mature national digital identity infrastructure (linked to Chinese national ID) that does not exist in the same form in Pakistan, making its verification approach difficult to replicate without a locally adapted solution like CNIC OCR and SIM-based OTP.
- No open API ecosystem or adaptable framework for deployment in other markets.

Summary: DiDi Carpool is the most philosophically aligned existing system to SmartPool, sharing the fundamental principle of peer-to-peer cost recovery rather than commercial profit. Its post-incident safety reforms, particularly mandatory facial recognition, also reflect a direction that SmartPool adopts proactively through CNIC-based biometric verification from day one. However, DiDi Carpool is entirely built around China's national identity infrastructure, payment ecosystem, and regulatory environment, none of which are applicable to Pakistan. SmartPool's contribution beyond DiDi lies in its Genetic Algorithm-based Virtual Meeting Point optimization for minimizing driver detour dynamically, its locally adapted verification pipeline using CNIC OCR and SIM-based OTP, its integration with Pakistani mobile wallets, and its gamified carbon credit ledger — all purpose-built for the realities of Karachi's commuting landscape rather than transplanted from a foreign market context.

5- Commutifi (Canada)

Introduction:

Commutifi is a Canada-based smart commute management platform founded in 2017, designed to help organizations, transit agencies, and city planners reduce single-occupancy vehicle usage among their employees and citizens. Unlike the previous systems discussed, Commutifi is not a direct consumer-facing carpooling app but rather a B2B commute intelligence platform that analyzes commuter data, recommends optimal shared travel options, and provides employers with tools to incentivize sustainable commuting behavior among their workforce. It has been deployed by universities, corporations, and municipal transit authorities across North America and Europe.

Working Model:

Commutifi works by first collecting commuter profile data including home location, workplace, working hours, and travel preferences. Its AI engine then analyzes this data across an entire organization to identify groups of employees with compatible routes and schedules, recommending carpooling matches, transit options, or cycling routes accordingly. Employers can configure incentive programs through the platform, such as subsidies, rewards points, or parking benefits for employees who choose sustainable commuting options. The platform generates detailed analytics dashboards showing commute emissions, cost savings, and modal shift trends for organizational reporting. Integration with third-party calendar systems and HR platforms allows automated scheduling alignment for carpool matching.

Advantages:

- AI-driven commute analysis across large user groups identifies carpooling opportunities that individuals would not find on their own.
- Organizational incentive engine creates structured, employer-backed motivation for sustainable commuting rather than relying on individual goodwill.
- Detailed carbon emissions tracking and reporting provides measurable environmental impact data useful for corporate sustainability goals.
- Calendar and HR system integration enables highly accurate schedule-based matching beyond simple route overlap.
- Proven in institutional environments such as universities and large corporations, demonstrating viability in structured commuter communities.

Disadvantages:

- Entirely B2B in nature — it requires an organizational sponsor such as an employer or university to deploy and fund the platform, making it inaccessible to the general public commuting independently.
- Does not handle real-time dynamic matching, GPS tracking, fare calculation, or payment processing — it is a recommendation and analytics tool, not an end-to-end carpooling transaction platform.
- No identity verification mechanism for individual users beyond organizational email credentials, making it unsuitable for open public deployment in markets like Pakistan where institutional emails are not universally held.
- Carbon tracking is reporting-oriented rather than user-facing and gamified, meaning it informs organizations but does not directly motivate individual commuters through rewards.
- No support for local payment ecosystems, real-time route optimization, or anti-fraud mechanisms relevant to the Pakistani context.
- Requires stable organizational infrastructure and internet connectivity assumptions that may not hold in developing urban environments.

Summary: Commutifi represents the most analytically sophisticated approach among the systems reviewed, demonstrating that AI-powered commute optimization and carbon tracking can be effectively combined at an institutional scale. Its emissions analytics and incentive engine directly inform SmartPool's gamified carbon credit ledger concept, validating the idea that environmental rewards can meaningfully shift commuter behavior. However, Commutifi is fundamentally an organizational tool that cannot function without an institutional intermediary, excludes the general public, and provides no transactional carpooling infrastructure whatsoever. SmartPool addresses all of these gaps by delivering an end-to-end peer-to-peer platform open to any Karachi resident, combining Commutifi's spirit of sustainability incentivization with real-time AI route matching, CNIC-based public verification, live GPS tracking, automated fare splitting, and local wallet payment — making it a comprehensive solution rather than an advisory layer sitting on top of other services.

Chapter # 03

3.0 Introduction:

SmartPool is a mobile-first, AI-powered carpooling platform designed for urban commuters in cities such as Karachi, who currently face high fuel costs, severe traffic congestion, and a lack of secure, reliable shared-transport options. Unlike conventional ride-hailing services such as Uber or Careem, SmartPool dynamically matches drivers and passengers travelling along overlapping routes, using an AI-powered optimization engine to compute Virtual Meeting Points that minimize driver detours while maximizing convenience for passengers. The platform further distinguishes itself through a layered identity-verification pipeline (CNIC OCR, SIM-based OTP, and on-device facial matching), a behavioral trust-score system, real-time GPS-based anti-fraud monitoring, an automated dynamic fare-splitting engine, and a gamified carbon-credit ledger that rewards sustainable travel habits.

3.1 Requirements Identifying Techniques

The following techniques were used by the team, in consultation with the project supervisor, to elicit and validate the requirements of SmartPool:

- **Stakeholder Interviews & Surveys:** Informal interviews and surveys were conducted with daily commuters, students, and working professionals in Karachi to understand their concerns around fuel cost, travel safety, and the "trust gap" that exists between strangers sharing a ride.
- **Competitor & Existing System Analysis:** Existing ride-hailing and carpooling applications were reviewed to identify functionality gaps, including the lack of real-time route flexibility, weak identity-verification mechanisms, and the absence of meaningful environmental incentives for users.
- **Document & Data Analysis:** Current fuel-price trends, local traffic-congestion patterns, and relevant data-protection considerations (e.g., handling of CNIC and financial information) were reviewed to define realistic, locally grounded constraints for the system.
- **Brainstorming Sessions:** Team and supervisor sessions were held to translate the project's problem statement into concrete, measurable functional capabilities, such as AI-based route matching, biometric verification, and dynamic fare splitting.
- **Scenario Walkthroughs:** Typical commuter scenarios (a driver posting a route, a passenger requesting a ride, multi-passenger fare splitting, identity verification at sign-up) were walked through step by step to surface functional and non-functional needs ahead of formal use case modeling.

3.2 Functional Requirements:

Table 3.1 Functional Requirement

Requirement 1	
Identifier	FR-01
Title	AI-Based Dynamic Route Matching & Virtual Meeting Point Generation
Requirement	The system shall use a Genetic Algorithm-based optimization engine, fed by the Google Maps Platform (Directions & Distance Matrix APIs), to match drivers and passengers travelling along overlapping routes and to compute optimal Virtual Meeting Points for pickup and drop-off.
Business Rule	A match shall only be suggested to a driver if the resulting detour does not exceed a predefined maximum threshold from their original route.
Requirement 2	
Identifier	FR-02
Title	CNIC-Based Identity Onboarding
Requirement	The system shall allow a user to register by scanning their CNIC, automatically extracting their personal details using OCR, and confirming their mobile number through a SIM-based OTP, without requiring an institutional email address.
Business Rule	A user account shall remain in “unverified” status, with ride-booking and driving privileges disabled, until both CNIC OCR extraction and OTP verification are successfully completed.

Requirement 3	
Identifier	FR-03
Title	Biometric Facial Verification
Requirement	The system shall capture a live selfie at registration and use on-device facial-geometry matching (Google ML Kit) to compare it against the photo extracted from the user's CNIC, in order to prevent identity theft.
Business Rule	Registration shall be automatically rejected, and flagged for manual review, if the live-selfie-to-CNIC-photo match confidence score falls below the defined security threshold.
Requirement 4	
Identifier	FR-04
Title	Real-Time GPS Tracking with Map Snapping
Requirement	The system shall continuously track driver and passenger device locations during an active ride and snap raw GPS coordinates to the nearest road segment to prevent GPS-drift errors in tracked distance.
Business Rule	Trip distance used for fare calculation shall always be derived from road-snapped coordinates and never from unprocessed raw GPS readings.
Requirement 5	
Identifier	FR-05
Title	Dual-Device Proximity Validation
Requirement	The system shall cross-reference the driver's and passenger's device locations at the time of pickup and drop-off to confirm that both parties were physically

	present together, preventing mileage fraud.
Business Rule	A ride shall not be marked as “completed” unless proximity validation confirms the driver and at least one passenger device were within an acceptable distance range at both pickup and drop-off.
Requirement 6	
Identifier	FR-06
Title	Dynamic Fare Calculation & Cost Splitting
Requirement	The system shall automatically calculate total trip cost from real-time fuel price, trip distance, a maintenance margin, and a base platform fee, and shall split this cost proportionally among the driver and passengers, including segment-based splitting for differing pickup/drop-off points.
Business Rule	The driver’s contribution to the total fare shall never be reduced to zero, and no passenger shall be charged for a route segment they did not travel.
Requirement 7	
Identifier	FR-07
Title	Secure Payment & Escrow Processing
Requirement	The system shall process passenger payments through an automated escrow mechanism using Easypaisa, JazzCash, or card payment, releasing funds to the driver only after the trip is marked complete.
Business Rule	Funds held in escrow shall be automatically refunded to the passenger if a ride is cancelled by the driver before it commences.

Requirement 8	
Identifier	FR-08
Title	Gamified Carbon-Credit & Trust-Score Tracking
Requirement	The system shall calculate and award carbon credits to users based on shared trips and estimated CO2 emissions saved, and shall maintain a behavioral trust score for each user derived from ride history and review sentiment analysis (NLTK/TextBlob).
Business Rule	A user’s trust score shall be recalculated after every completed ride and shall directly influence their visibility and ranking within the AI-based matching engine.

Non - Functional Requirements:

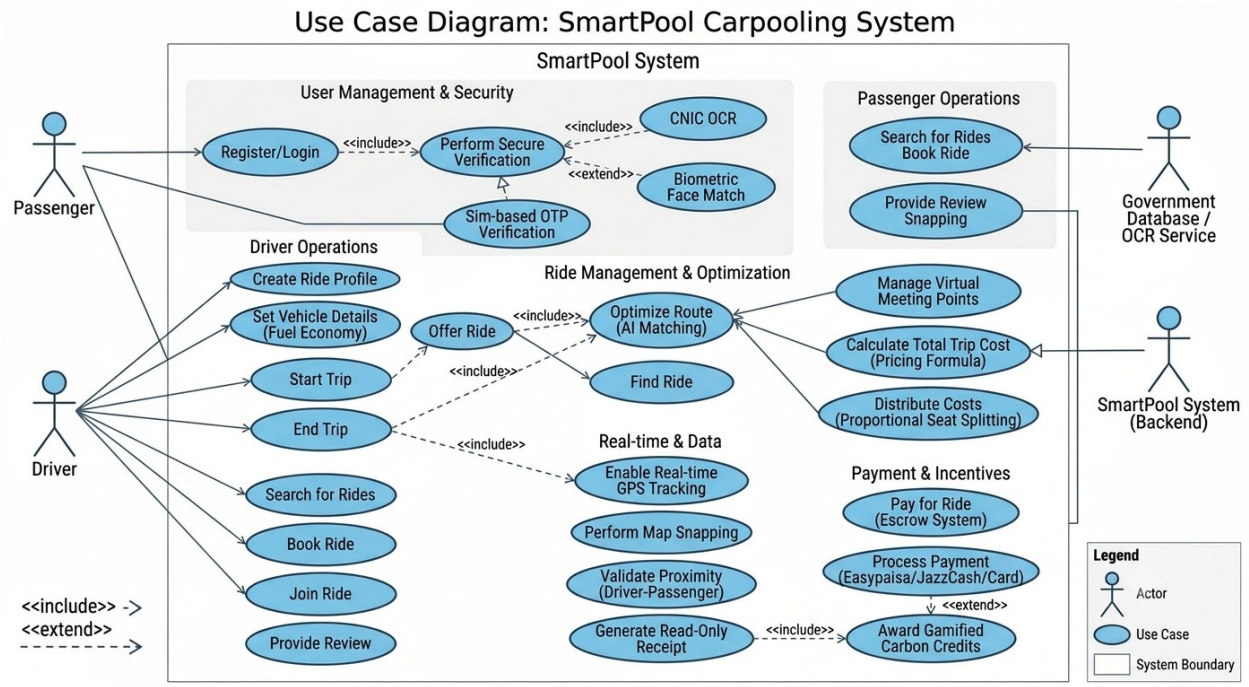
In addition to the functional capabilities listed in Table 3.1, SmartPool must satisfy the following quality attributes to operate safely and reliably as a public, AI-driven carpooling platform in Karachi.

Table 3.2 Non-Functional Requirements

Category	Description
Performance	Route-matching and Virtual Meeting Point calculations shall return results within a few seconds of a request being submitted, even under typical city-wide traffic conditions.
Security	All CNIC images, biometric facial data, and payment information shall be encrypted in transit and at rest; ride and payment records shall become read-only once a trip is completed, preventing post-hoc tampering.
Reliability	Automated fare-split calculations shall be applied consistently for every ride, and proximity/road-snapping validation shall reliably detect and flag GPS spoofing or mileage-fraud attempts.
Usability	The map-based mobile interface shall allow a first-time user to complete registration, identity verification, and ride request/offer in a small number of simple, guided steps without needing technical assistance.
Scalability	The backend (Node.js with PostgreSQL/PostGIS) shall be designed to scale horizontally so that matching speed does not noticeably degrade as the number of concurrent users and ride requests across Karachi grows.
Availability	Core ride-matching and payment services shall remain available with minimal downtime, and shall degrade gracefully (e.g., cached map data, queued requests) if third-party APIs such as Google Maps or the payment gateway become temporarily unavailable.
Energy Efficiency	Continuous GPS tracking and on-device AI processing (facial matching, road snapping) shall be optimized to keep battery consumption on users' mobile devices within an acceptable range during an active ride.
Portability / Compatibility	The application shall be built on a cross-platform framework (Flutter/React Native) so that it runs consistently across the range of Android devices commonly used by

	Karachi commuters.
Maintainability	The system shall follow a modular, service-oriented architecture (separate matching, verification, and payment services) so that individual components can be updated, fixed, or extended independently during and after the Agile/Scrum development cycle.

3.4 Use Case Diagram:



3.4.1 Explanation of Use Case Diagram:

Overview

This UML Use Case Diagram outlines the functional requirements of the SmartPool Carpooling System. It illustrates the interactions between external actors (users/systems) and the internal functionalities (use cases) of the platform, grouped into core operational modules.

Core Actors

- **Passenger:** A user who searches for, books, and pays for rides.
- **Driver:** A user who registers vehicles, offers rides, and manages trips.
- **Government Database / OCR Service:** An external secondary actor used to verify identities (via CNIC and facial data).
- **SmartPool System (Backend):** The automated backend actor that handles complex logic like route optimization, cost distribution, and trip calculations.

Key Operational Modules

1. User Management & Security

- Handles **Register/Login** functionality.
- Triggers **Perform Secure Verification** (via `<<include>>`), which mandatorily uses **CNIC OCR** and **Sim-based OTP Verification**.
- **Biometric Face Match** conditionally extends (`<<extend>>`) this verification process for higher security.

2. Driver & Passenger Operations

- **Driver Operations:** Allows drivers to create ride profiles, input vehicle details, start/end trips, and look for rides.
- **Passenger Operations:** Enables passengers to search for and book rides, as well as provide reviews.
- **Offer Ride** automatically includes (`<<include>>`) the **Optimize Route (AI Matching)** and **Find Ride** functions to pair drivers with optimal passengers.
- **End Trip** automatically includes (`<<include>>`) **Enable Real-time GPS Tracking** data finalization.

3. Ride Management & Optimization

- Managed heavily by the **SmartPool Backend**.
- Includes background processes like **Managing Virtual Meeting Points**, **Calculating Total Trip Costs** (via pricing formulas), and **Distributing Costs** based on proportional seat splitting.

4. Real-time & Data

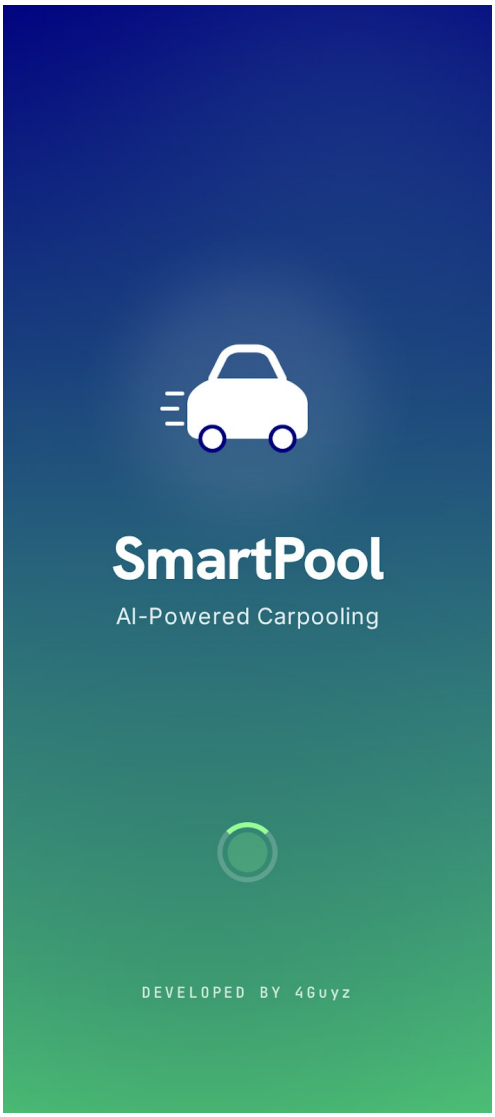
- Handles ongoing trip mechanics including **Real-time GPS Tracking**, **Map Snapping** (aligning GPS coordinates to roads), and **Validating Proximity** between driver and passenger.
- Generates a **Read-Only Receipt** at trip completion, which automatically includes (`<<include>>`) the logic to **Award Gamified Carbon Credits**.

5. Payment & Incentives

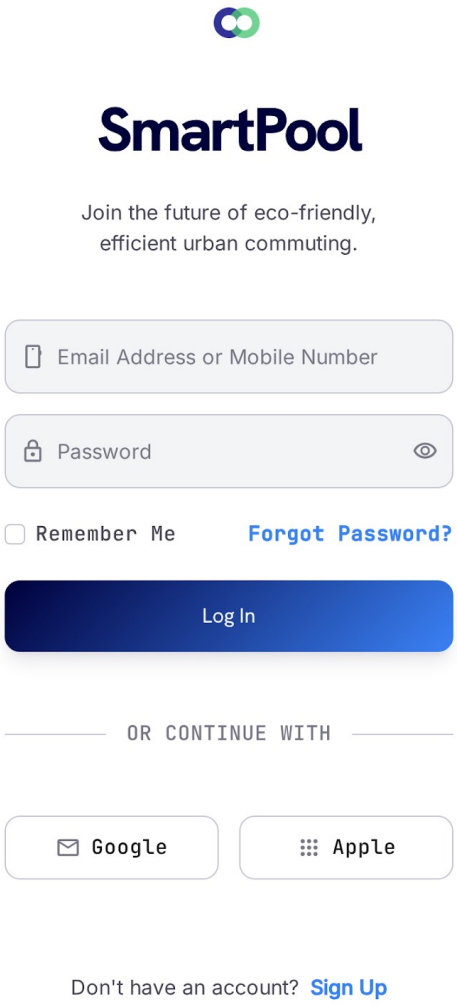
- Facilitates transactions using a secure **Escrow System** (**Pay for Ride**).
- Processes payments via regional channels (**Easypaisa/JazzCash/Card**).

3.5 Prototype:

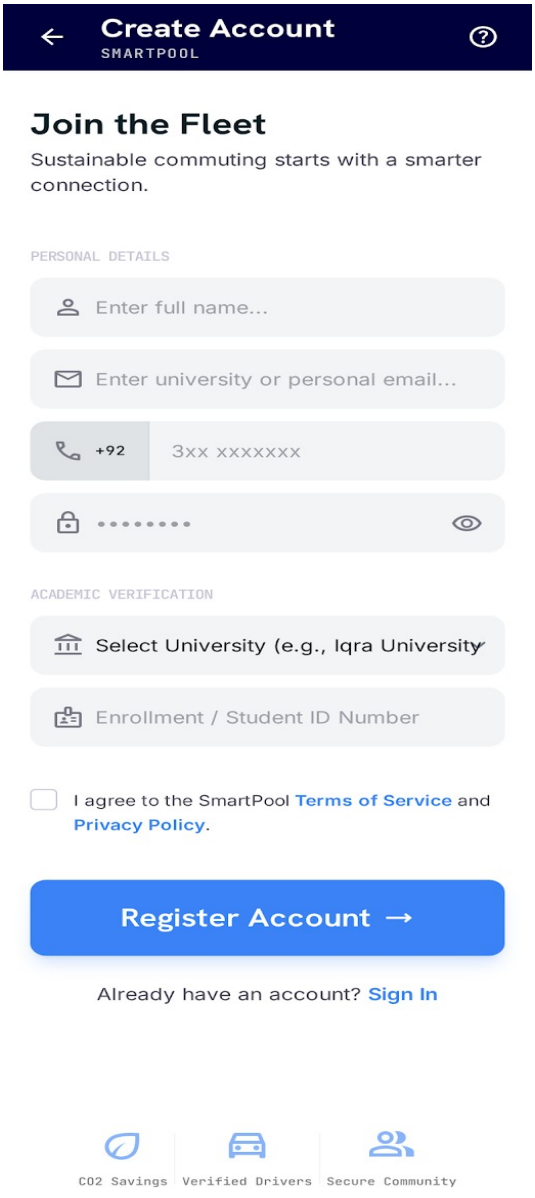
Splash Screen



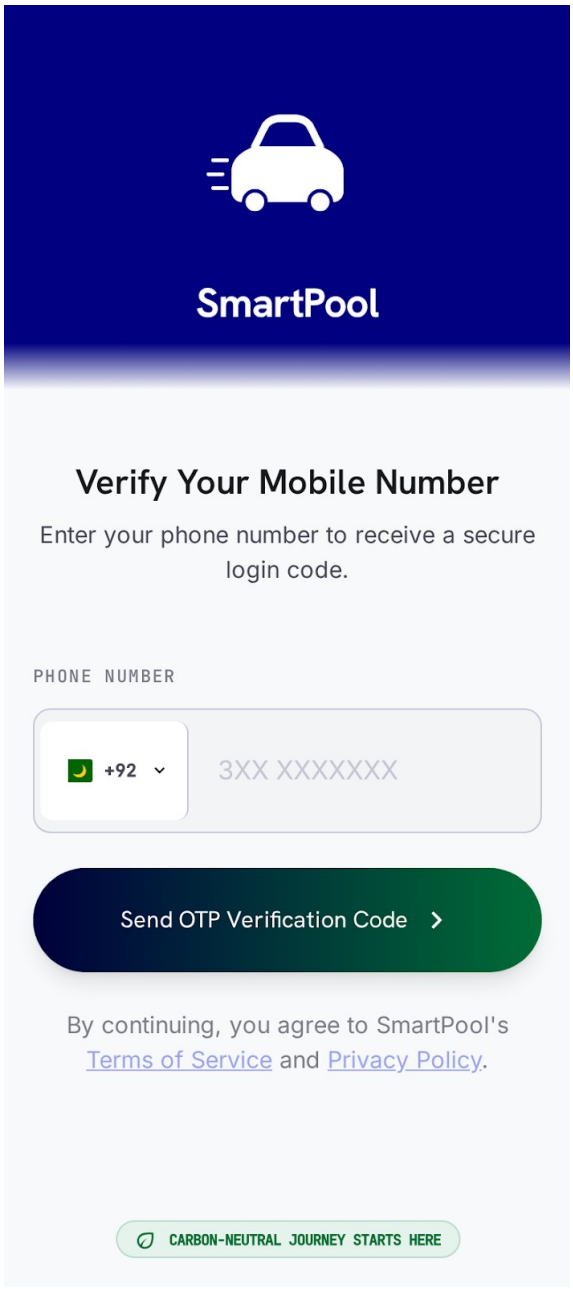
Login Screen



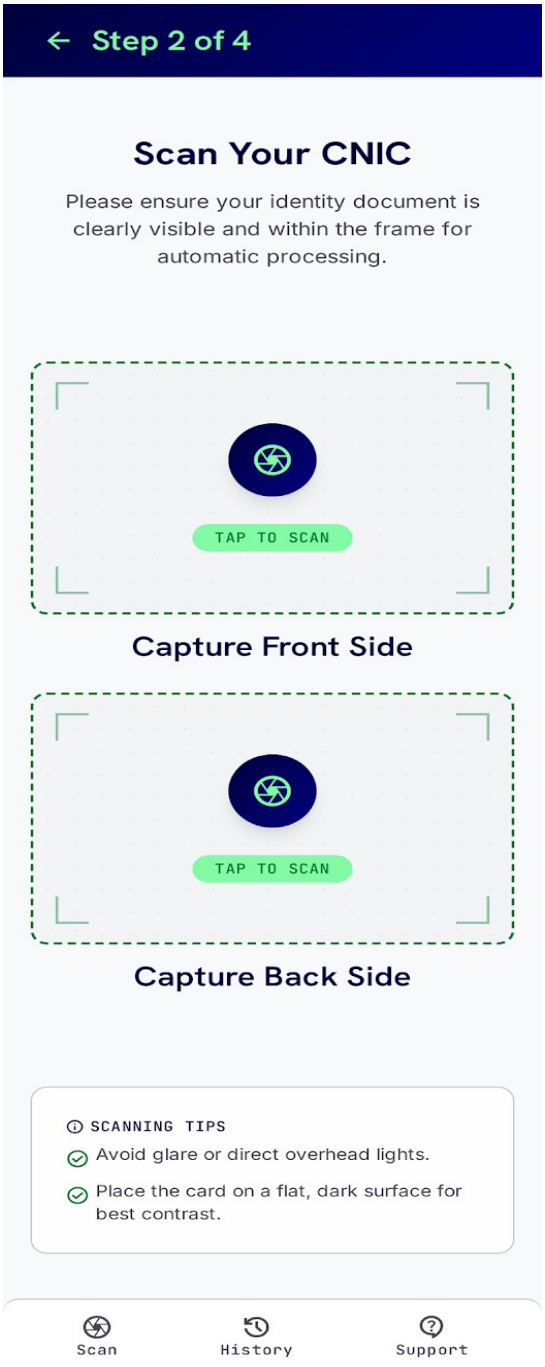
Signup Screen



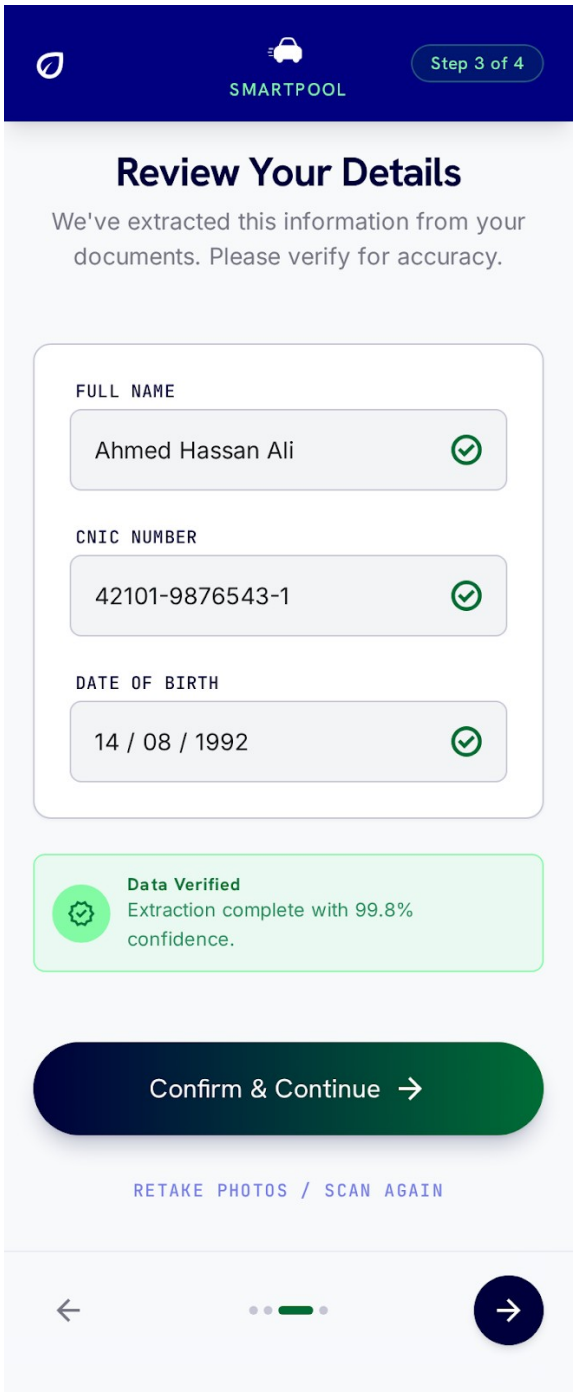
Phone Registration



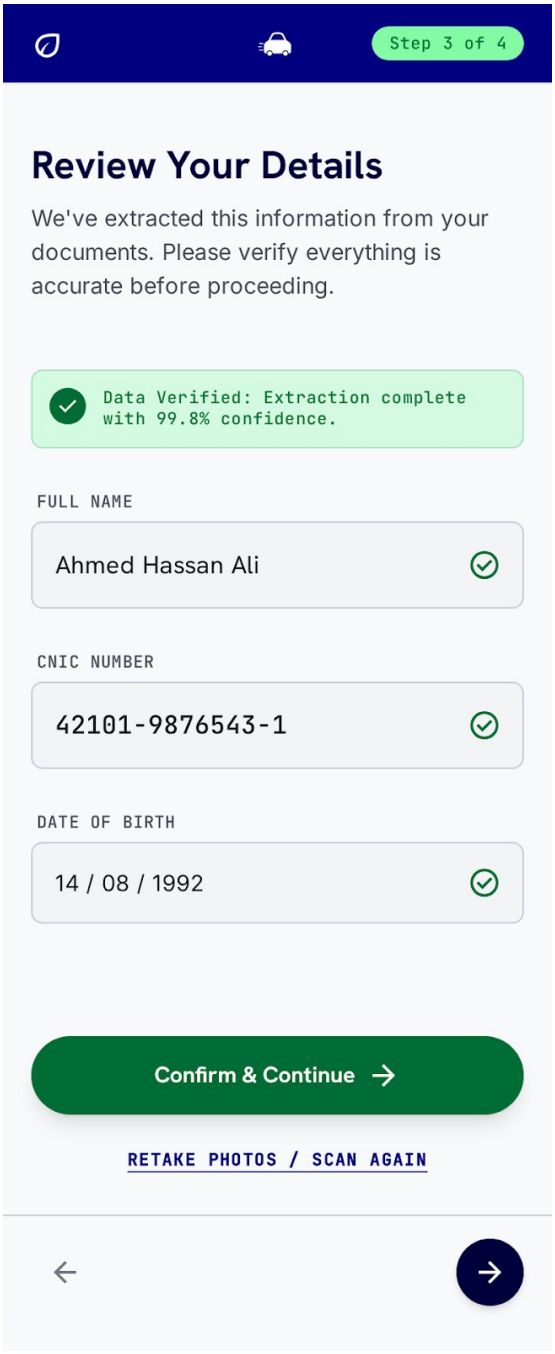
CNIC Identity Scanning



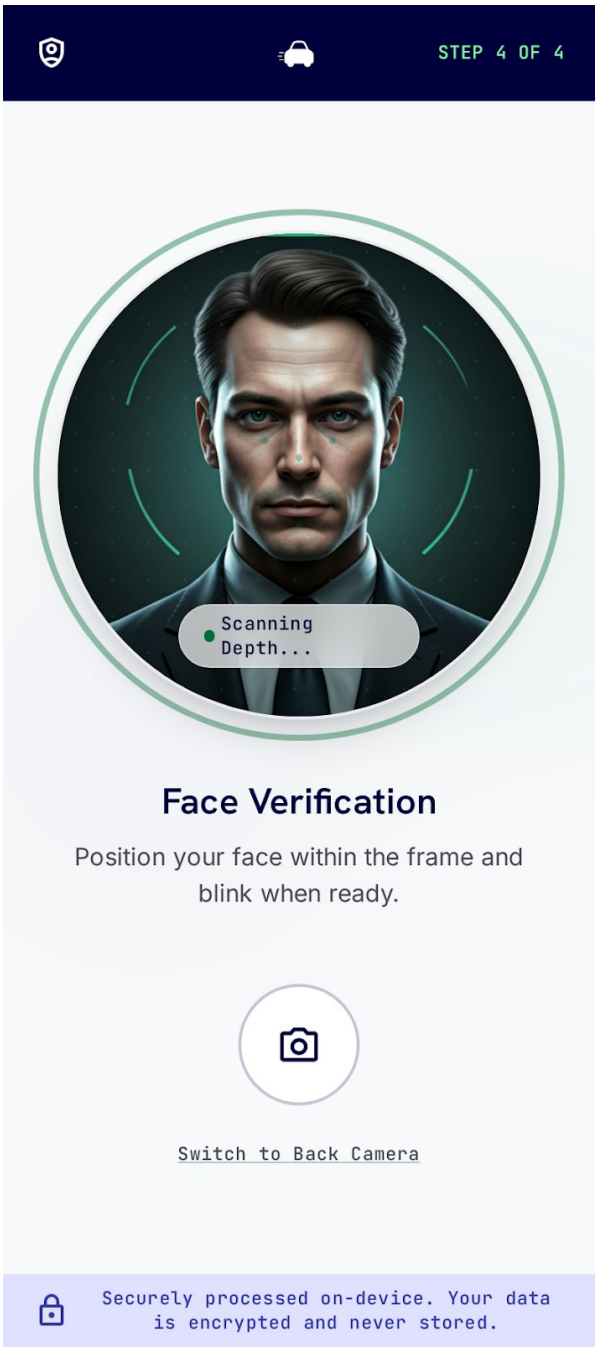
Data Review & Confirmation



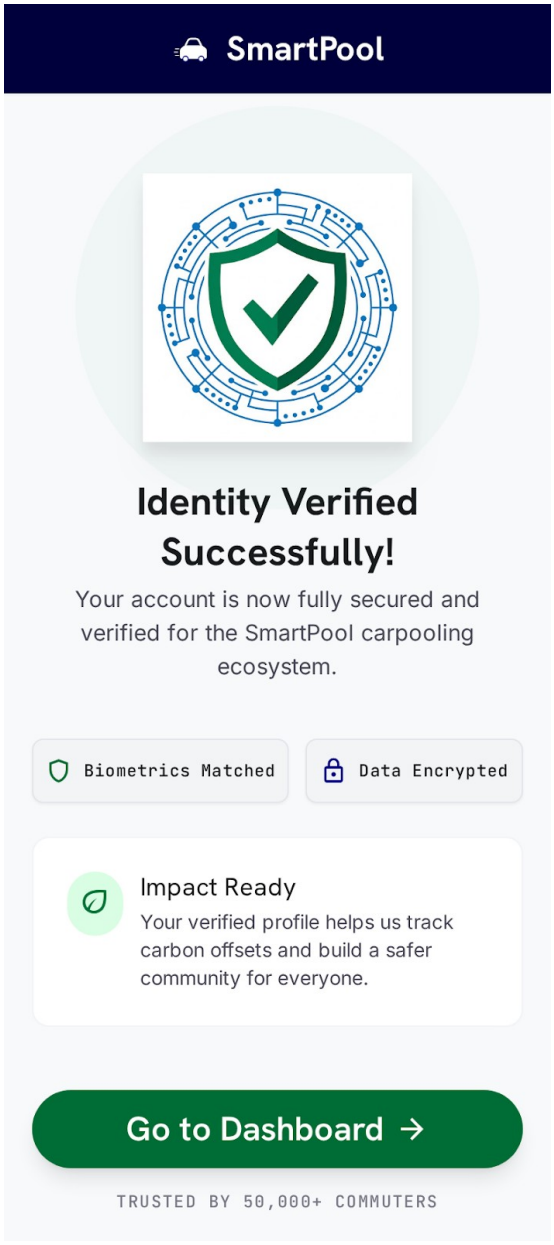
Data Review & Confirmation



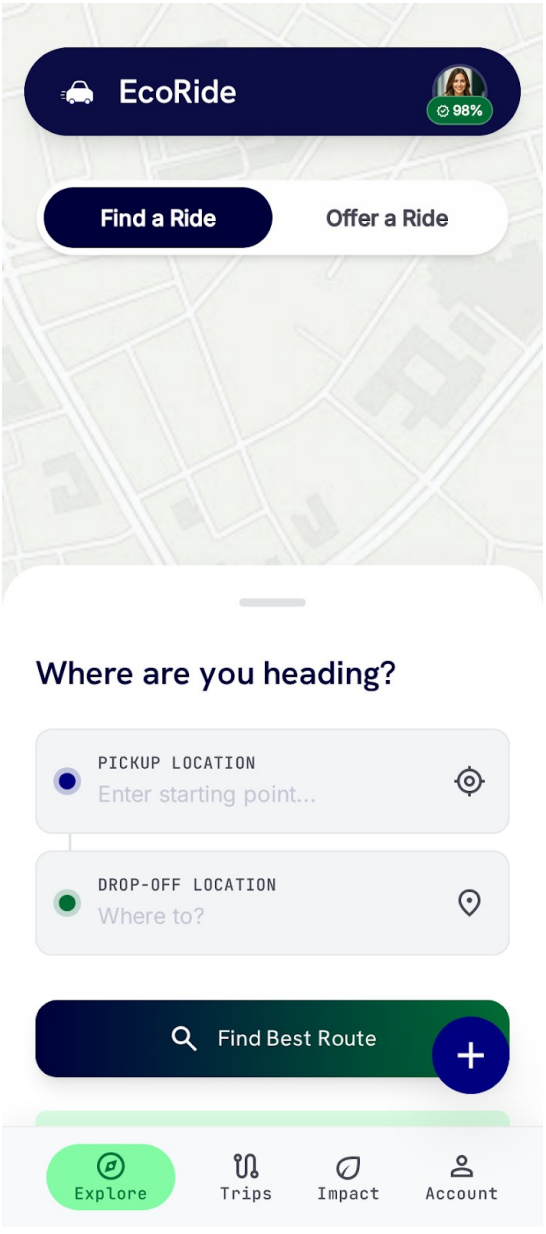
Facial Biometric Capture



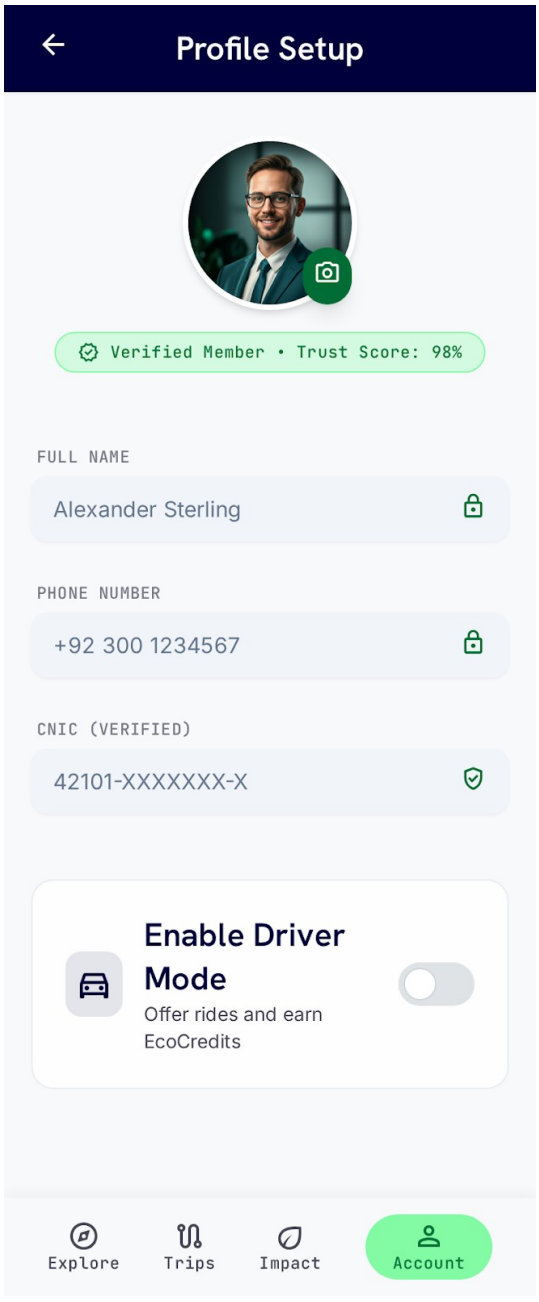
Identity Verification Success



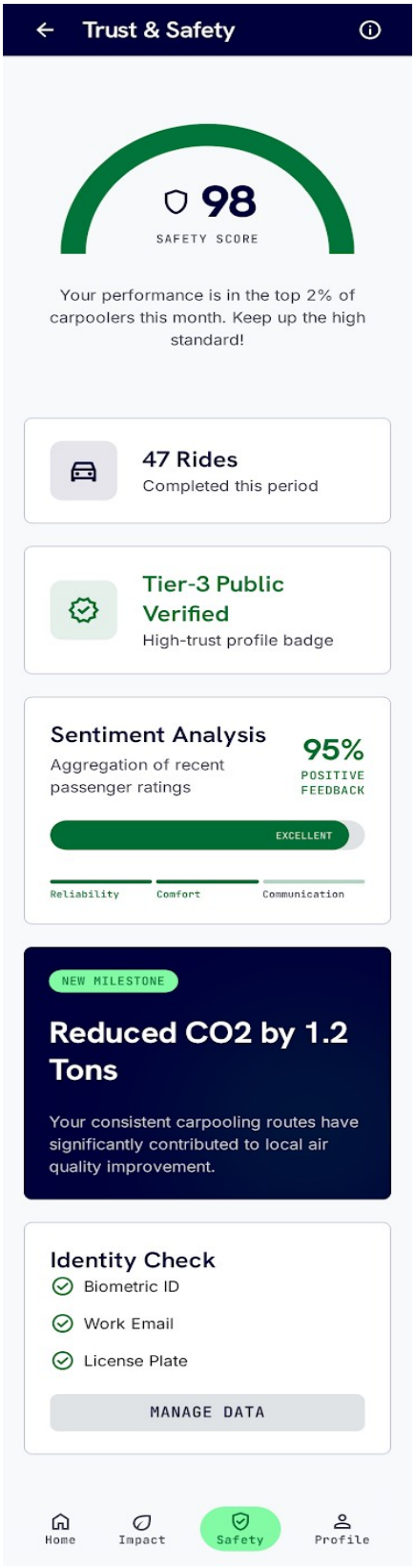
Home Dashboard



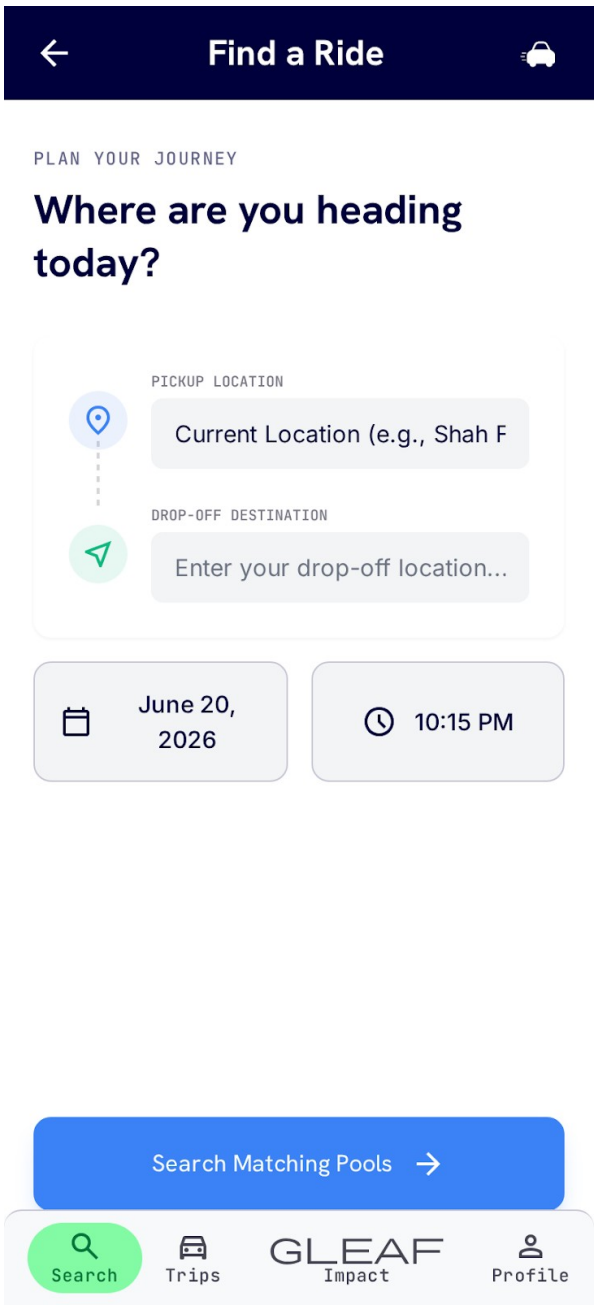
Profile & Vehicle Setup



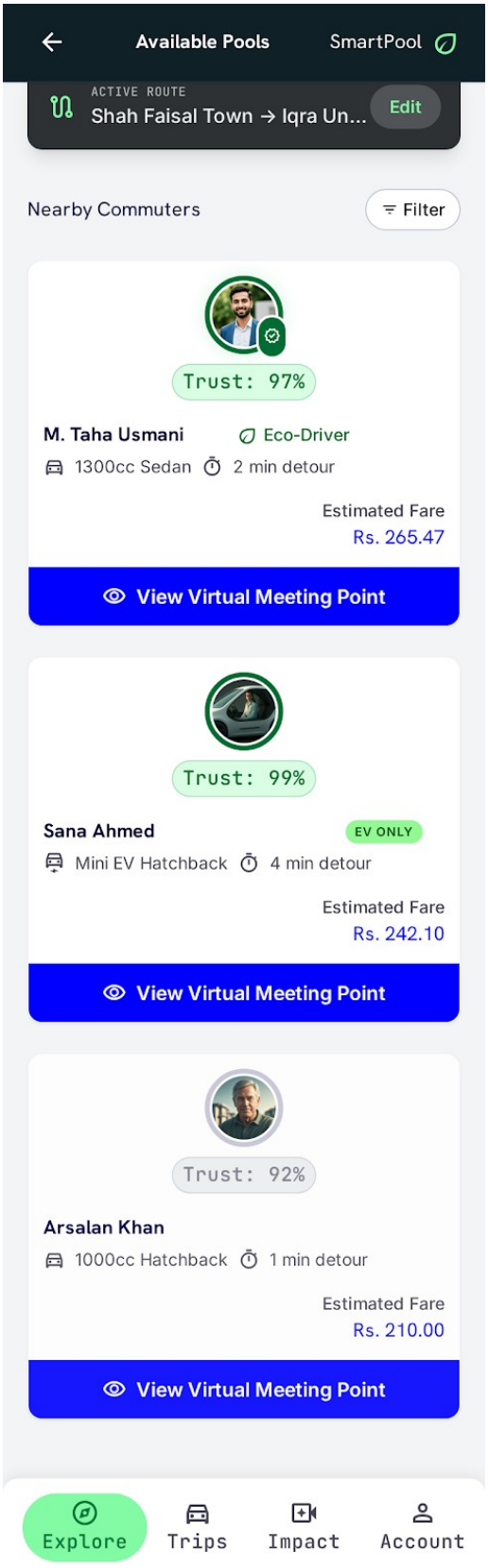
Trust & Safety Analytics



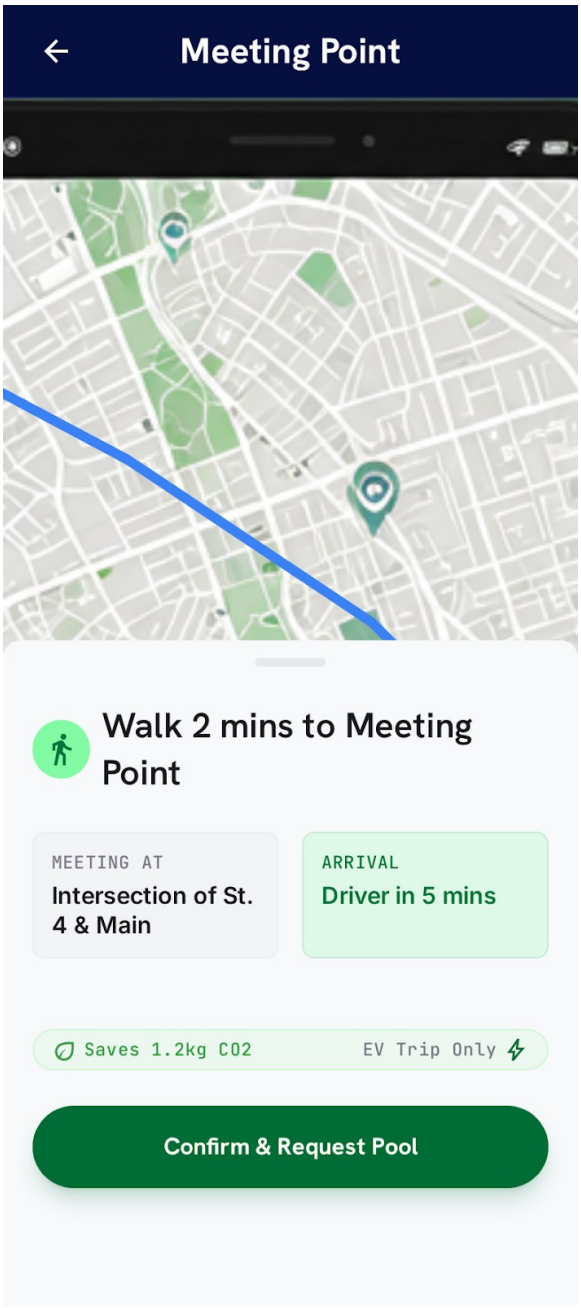
Find a Ride



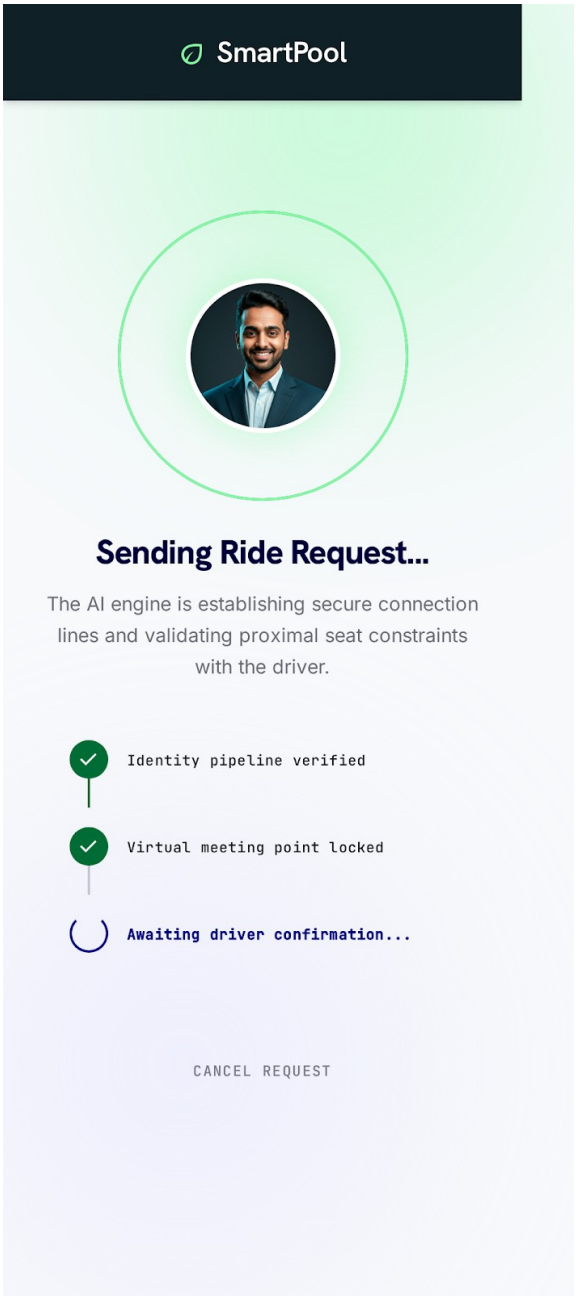
Available Pools Search Result



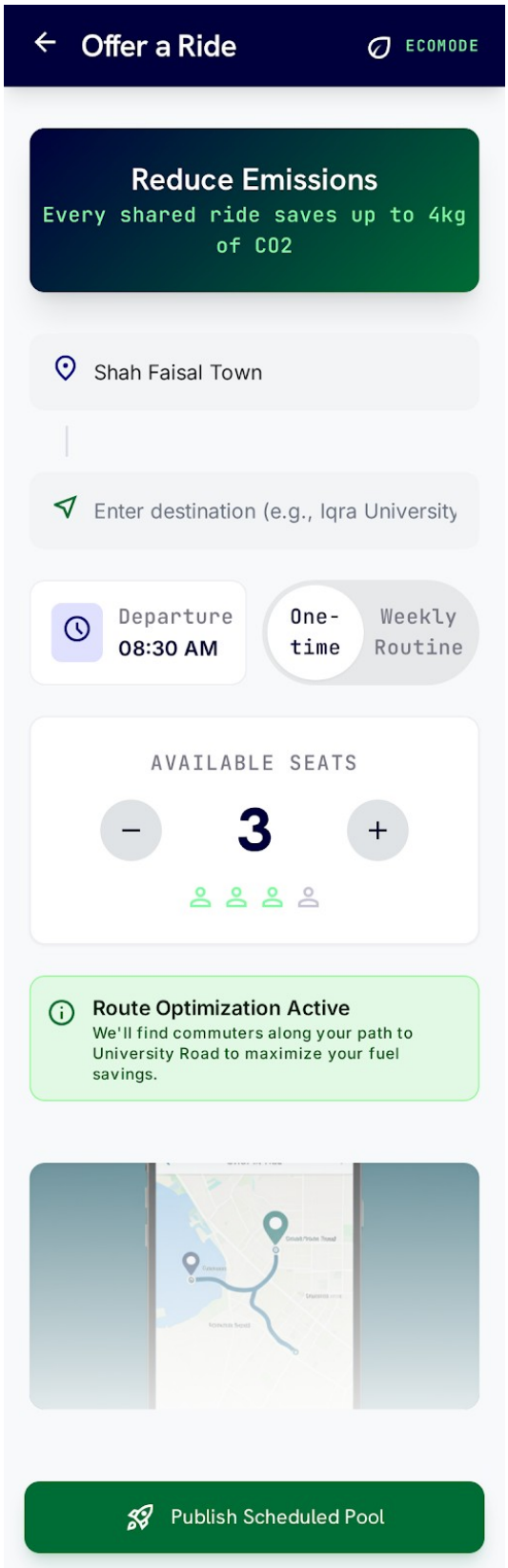
Meeting Point Discovery



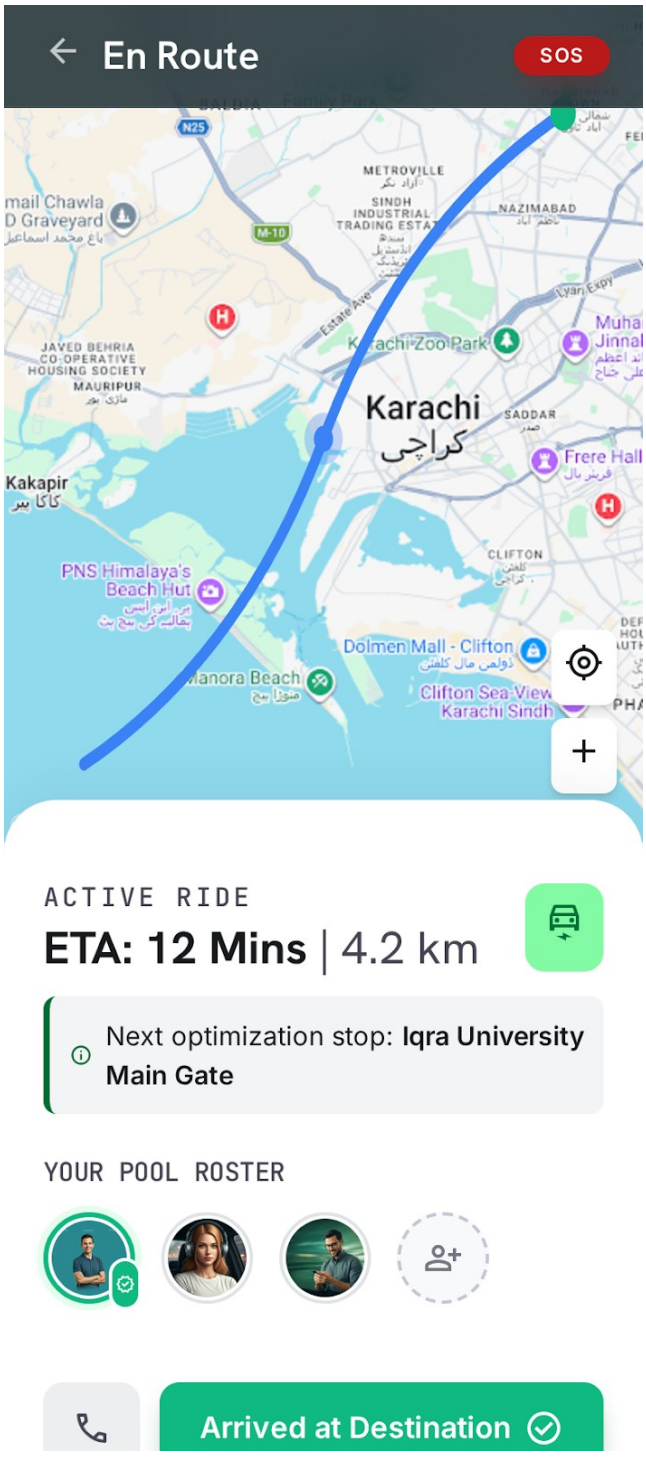
Booking Confirmation Status



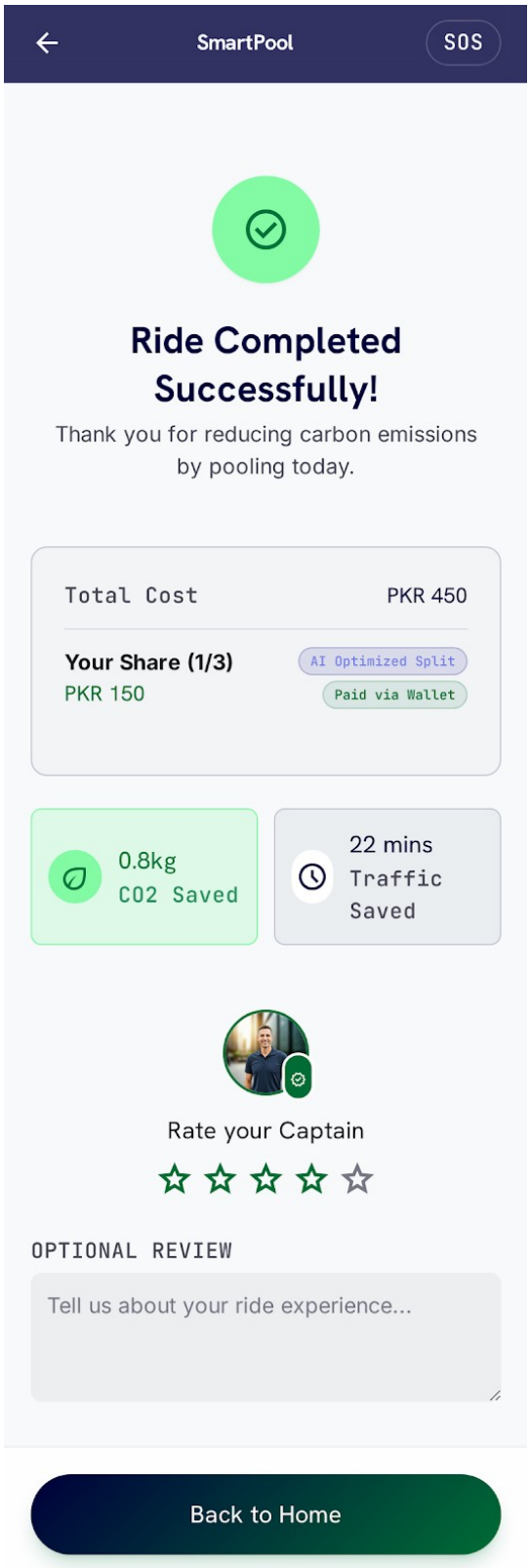
Offer a Ride



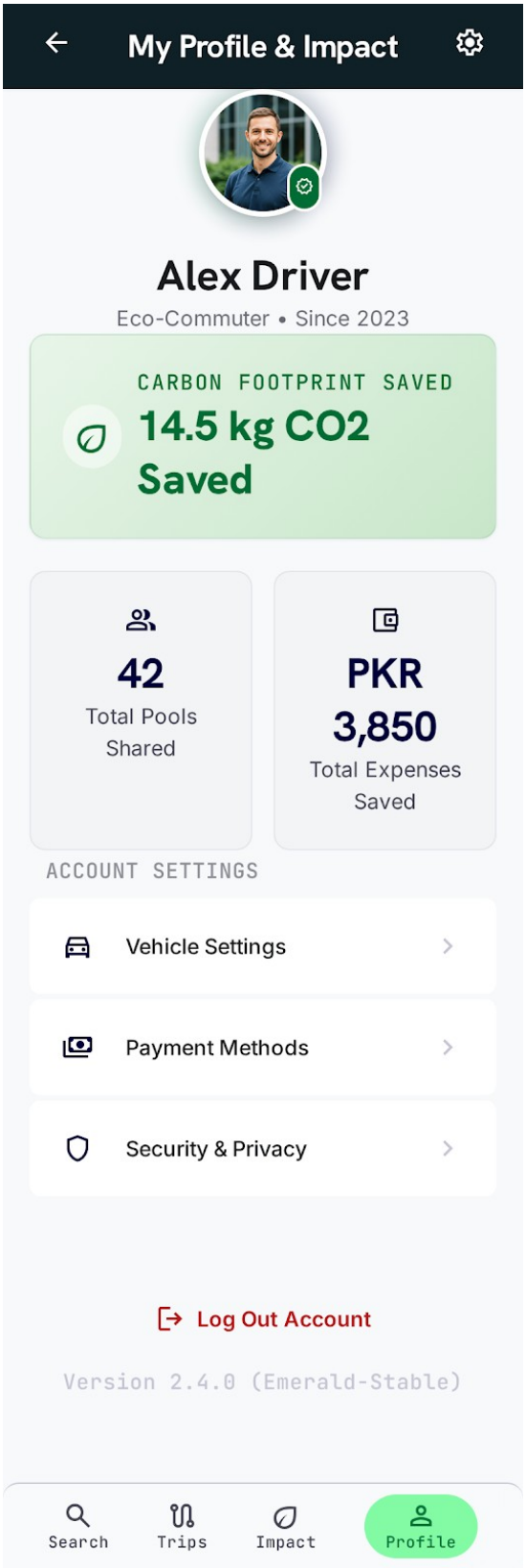
Active Ride Tracking & Navigation



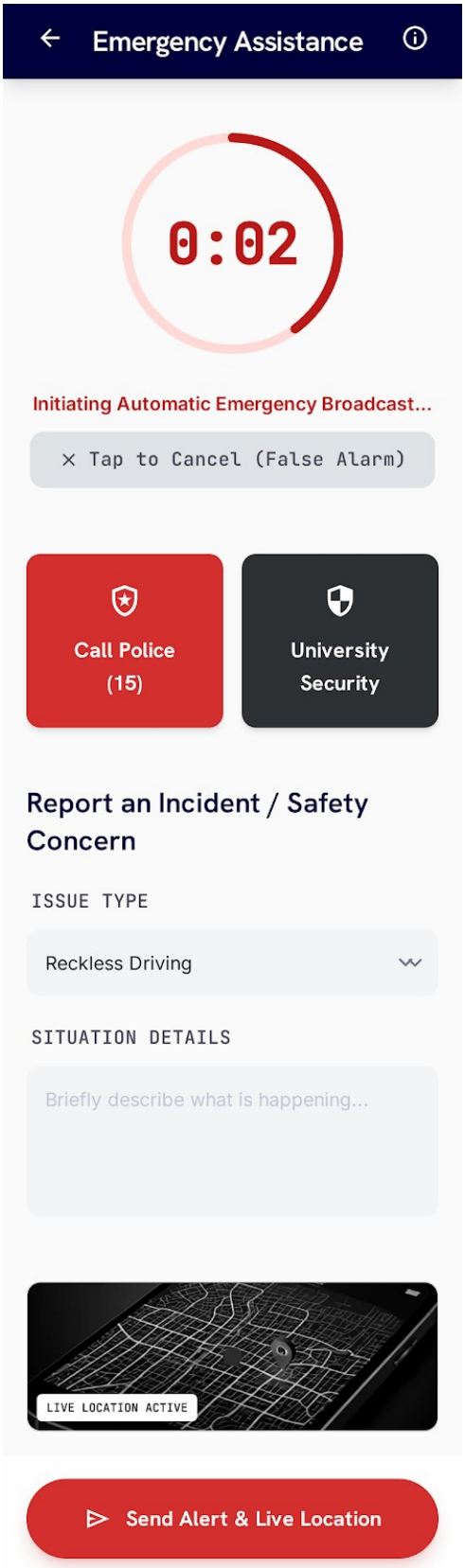
Ride Completion & Fare Split Summary



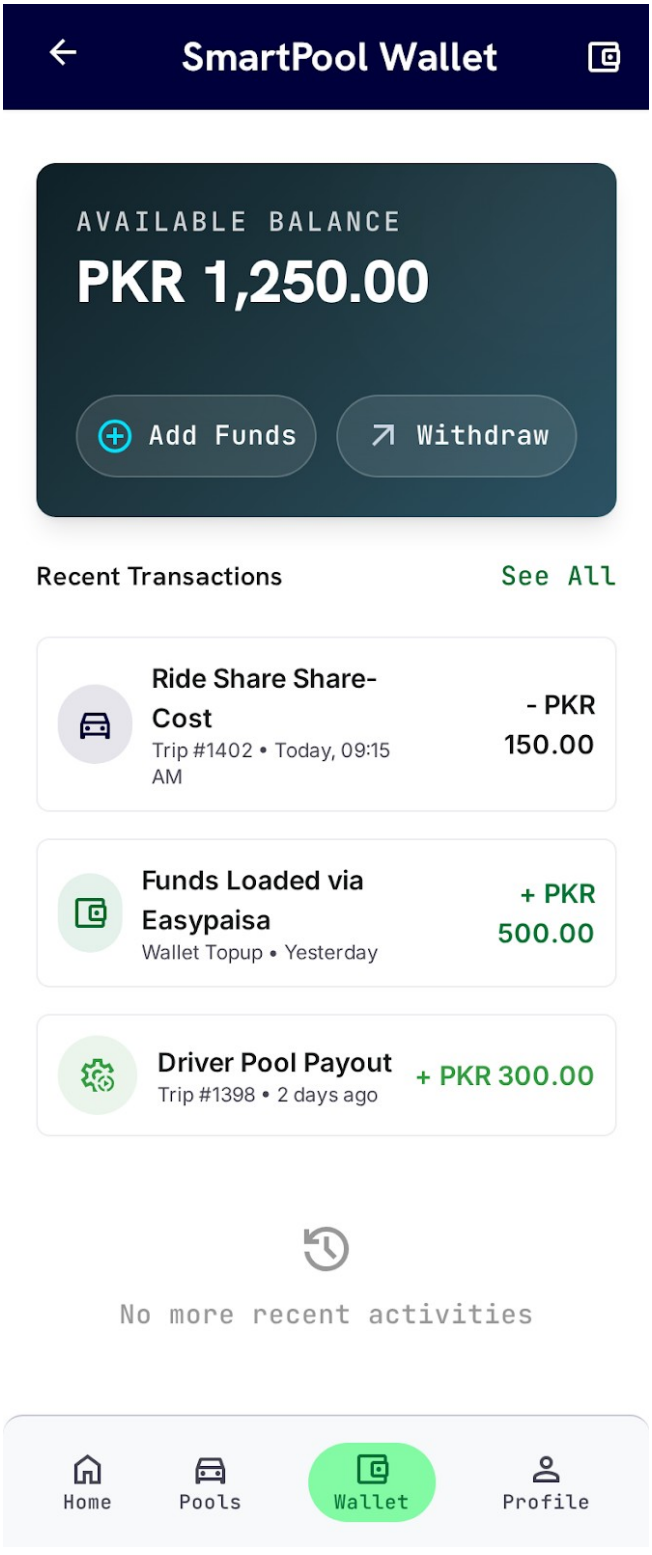
My Profile & Impact



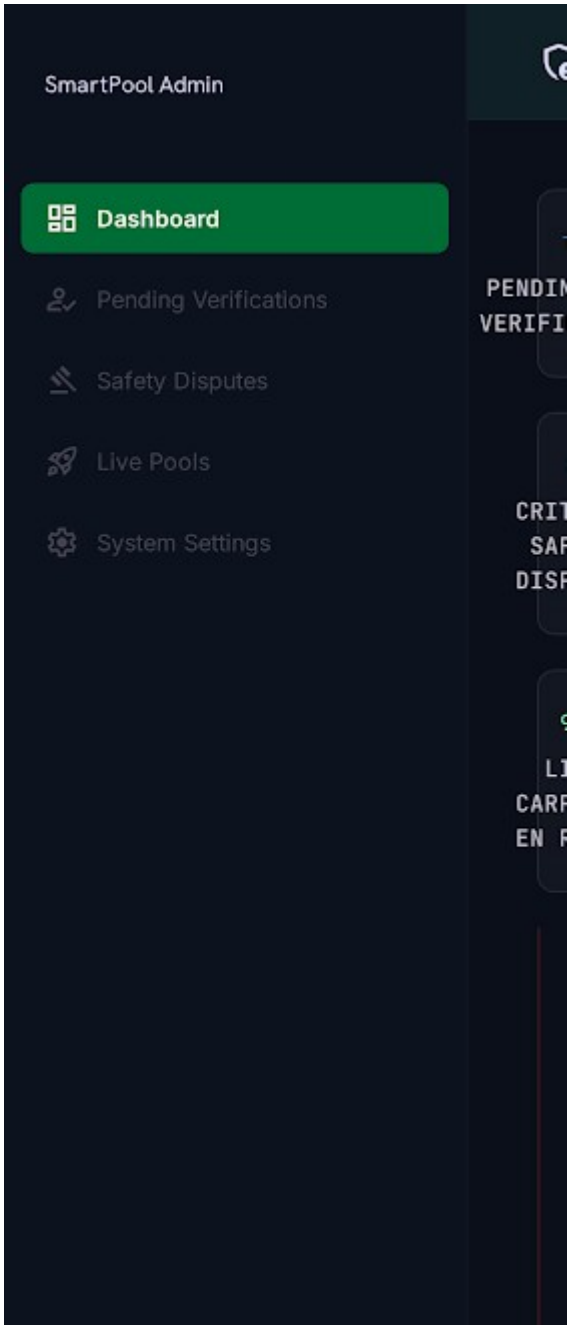
Emergency Assistance & SOS



SmartPool Wallet & Transaction History



SmartPool System Control Unit Dashboard



Chapter # 04

4.0 Introduction

This budget document for the SmartPool Smart Carpooling System outlines the complete financial distribution required for the successful implementation and public launch of the project. It includes detailed heads of expenditure covering Technical HR Deployment, Support Staff, Equipment, Software & Cloud Services, Play Store Launch & App Distribution, Miscellaneous expenses, Audit Charges, and a Contingency Fund. As SmartPool is intended for public deployment on the Google Play Store, the budget also accounts for all associated app publishing, listing, and initial user acquisition costs. The estimated total budget for this project is Rs. 2,175,000. This budget is designed to cover all development, staffing, infrastructure, and operational requirements necessary to deliver a secure, AI-powered, and publicly accessible carpooling platform for urban commuters in Karachi.

4.1 Heads of Expenditure

Sr.	Description	Amount (Rs)
1	Technical HR Deployment Cost	Rs. 1,080,000
2	Support Staff	Rs. 180,000
3	Equipment	Rs. 200,000
4	Software & Cloud Services	Rs. 345,000
5	Play Store Launch & App Distribution	Rs. 45,000
6	Miscellaneous	Rs. 150,000
	Total Sub Amount	**Rs. 2,000,000**

4.1.1 Technical HR Deployment Cost — Rs. 1,080,000

SmartPool requires four specialized developers working across its distinct technical subsystems: the Flutter/React Native frontend, the Node.js backend, the AI/ML optimization engine, and the security and verification pipeline. Each team member is allocated a stipend of Rs. 30,000 per month over the 9-month active development period (Weeks 17–30).

Sr.	Role / Resource	Amount (Rs)
1	Full-Stack Developer (Flutter/React Native + Node.js) — 1 member × 9 months × Rs. 30,000/month	Rs. 270,000
2	Backend & Database Developer (Node.js + PostgreSQL/PostGIS) — 1 member × 9 months × Rs. 30,000/month	Rs. 270,000
3	AI/ML Engineer (Genetic Algorithm + Python FastAPI + NLP) — 1 member × 9 months × Rs. 30,000/month	Rs. 270,000
4	Security & Verification Engineer (Google ML Kit + Escrow + Anti-Spoofing) — 1 member × 9 months × Rs. 30,000/month	Rs. 270,000
	Technical HR Sub-Total	**Rs. 1,080,000**

4.1.2 Support Staff — Rs. 180,000

Support staff costs cover non-core but essential roles including the academic supervisor providing advisory oversight, a quality assurance coordinator managing testing cycles, and a technical documentation and report writer responsible for SDLC documentation and final FYP thesis compilation.

Sr.	Role / Resource	Amount (Rs)
1	Project Supervisor / Academic Coordinator (advisory sessions)	Rs. 60,000
2	Quality Assurance & Testing Coordinator (UAT, integration testing cycles)	Rs. 60,000
3	Technical Documentation & Report Writer	Rs. 60,000
	Support Staff Sub-Total	**Rs. 180,000**

4.1.3 Equipment — Rs. 200,000

Hardware requirements are minimal as SmartPool is a software-based mobile platform. A primary development laptop is required to host the full software stack, and a physical Android device is essential for testing GPS tracking accuracy, Google ML Kit biometric matching, OTP flows, and Easypaisa/JazzCash wallet integration.

Sr.	Item	Amount (Rs)
1	Development Laptop (host Flutter SDK, Node.js, PostGIS, Genetic Algorithm training)	Rs. 165,000
2	Android Test Device (physical GPS, ML Kit biometric, OTP, and wallet testing)	Rs. 35,000
	Equipment Sub-Total	**Rs. 200,000**

4.1.4 Software & Cloud Services — Rs. 345,000

Cloud and software costs cover all third-party APIs, hosting infrastructure, development tools, and service licenses required for the 12-month project lifecycle. This includes the Google Maps Platform for route optimization data, Firebase for real-time features, SMS OTP gateway for SIM-based user verification, payment gateway merchant fees for Easypaisa/JazzCash, and all supporting developer and design tools.

Sr.	Service / License	Amount (Rs)
1	VPS Server Hosting — Node.js API & Python FastAPI microservices (12 months)	Rs. 24,000
2	Managed PostgreSQL/PostGIS Database Instance (12 months)	Rs. 18,000
3	Google Maps Platform API — Directions & Distance Matrix (production quota)	Rs. 15,000
4	Firebase (Realtime DB, push notifications, analytics — 12 months)	Rs. 18,000
5	SMS/OTP Gateway — SIM-based registration OTP tokens (commercial volume)	Rs. 10,000
6	SSL/TLS Certificate — production-grade API encryption for wallet transactions	Rs. 5,000
7	Easypaisa/JazzCash SDK Integration & Merchant	Rs. 25,000

	Registration Fees	
8	GitHub Team Plan — version control and CI/CD pipeline (12 months)	Rs. 15,000
9	Domain Name Registration (12 months)	Rs. 5,000
10	Figma Professional — UI/UX prototyping and high-fidelity mockups (12 months)	Rs. 20,000
11	FastAPI Hosting for Carbon Offset Microservice (12 months)	Rs. 15,000
12	Cloud Backup & Storage (ride records, biometric logs — read-only constraints)	Rs. 15,000
13	Postman API Testing Suite & Load Testing Tools	Rs. 10,000
14	Google ML Kit / TFLite Model Hosting & Testing Environment	Rs. 10,000
15	Miscellaneous Software Licenses & Developer Tools	Rs. 80,000
	Software & Cloud Sub-Total	**Rs. 345,000**

4.1.5 Play Store Launch & App Distribution — Rs. 45,000

Since SmartPool will be publicly deployed on the Google Play Store for download by all Karachi commuters, this cost head covers the one-time developer account registration, app signing, store listing asset creation, closed beta testing setup, App Store Optimization (ASO) for discoverability, and an initial launch marketing campaign to drive first-wave user acquisition.

Sr.	Item	Amount (Rs)
1	Google Play Store Developer Account Registration (one-time fee ~ $25 USD)	Rs. 7,000
2	App Signing Certificate & Keystore Management	Rs. 3,000
3	Play Store Listing Assets — screenshots, promotional graphics, feature banner	Rs. 10,000
4	Beta Testing Distribution & Closed Testing Track Setup	Rs. 5,000
5	App Store Optimization (ASO) — metadata, keywords, localization for Pakistani market	Rs. 10,000
6	Initial User Acquisition & Launch Marketing (social media, referral campaigns)	Rs. 10,000
	Play Store & Launch Sub-Total	**Rs. 45,000**

4.2 Additional Charges

Description	Amount (Rs)
Audit Charges (FYP evaluation, documentation review, panel assessment)	Rs. 45,000
Contingency Fund (unforeseen technical expenses, API overages, emergency hardware)	Rs. 100,000
Institutional / Organizational Overheads (university resource usage, lab access)	Rs. 30,000

4.3 Total Budget

The total budget for the SmartPool — Smart Carpooling System project amounts to Rs. 2,175,000, calculated as follows:

Sub-Total (Heads of Expenditure): Rs. 2,000,000

Audit Charges: Rs. 45,000

Contingency Fund: Rs. 100,000

Institutional/Organizational Overheads: Rs. 30,000

TOTAL PROJECT BUDGET	**Rs. 2,175,000**

4.4 Funding Sources

Sr.	Funding Source	Amount (Rs)
1	Self-Funded (FYP Team Members — equal contribution)	Rs. 2,175,000
	Total Funding	**Rs. 2,175,000**

Chapter # 05

5.0 Problem

Urban commuters in Karachi face a convergence of three unresolved crises: extreme fuel costs that continue to rise, severe and worsening traffic congestion that inflates daily commute times, and a deep-rooted trust gap that prevents strangers from safely sharing vehicles. Existing commercial ride-hailing services such as Careem and Uber operate on a profit-based driver model, making them financially unsustainable for daily use. There is no local platform that enables genuine peer-to-peer cost-sharing carpooling while simultaneously addressing identity fraud, GPS mileage manipulation, and the absence of local digital payment integration. Citizens without institutional emails or formal employment are entirely excluded from what verification systems do exist. Furthermore, no current solution incentivizes environmentally responsible commuting behaviour, leaving the significant carbon reduction potential of carpooling entirely unrealised.

5.1 Solution

SmartPool is a mobile-first, AI-powered peer carpooling platform purpose-built for urban commuters in Karachi. Rather than connecting passengers with professional drivers for commercial gain, SmartPool enables ordinary vehicle owners commuting on regular routes to share their empty seats with passengers travelling the same direction, splitting only the actual fuel and maintenance costs among all occupants. The platform uses Genetic Algorithms to compute optimal Virtual Meeting Points that minimise driver detour while maximising passenger convenience. A tiered security pipeline using CNIC OCR, on-device biometric facial matching via Google ML Kit, and SIM-based OTP verification onboards any Pakistani citizen safely without requiring institutional credentials. GPS Map Snapping and Dual-Device Proximity Validation eliminate mileage fraud and verify physical co-presence. An automated escrow system through Easypaisa and JazzCash handles all payments securely. A gamified carbon-credit ledger rewards users for every sustainable ride completed.

5.2 Unique Value Proposition

SmartPool delivers the only end-to-end carpooling platform designed specifically for Karachi's commuting realities. It is the sole solution that combines AI-driven dynamic route optimization, CNIC-based biometric identity verification open to the general public, local mobile wallet escrow, GPS anti-spoofing, and environmental carbon-credit incentives within a single peer-to-peer cost-sharing system. Unlike commercial ride-hailing apps, SmartPool operates on a non-profit cost distribution model where drivers recover only their fuel and maintenance expenses — making daily shared commuting genuinely affordable. By removing the need for institutional email addresses and accepting any Pakistani CNIC as the identity anchor, SmartPool is accessible to the full spectrum of Karachi's 20+ million residents rather than a privileged subset.

5.3 Unfair Advantage

SmartPool's competitive moat is built on several layers that are extremely difficult to replicate quickly. Its custom Genetic Algorithm route optimization engine is trained specifically on Karachi road network data and local commute patterns, making it far more accurate than generic matching systems transplanted from foreign markets. The CNIC-based on-device biometric verification pipeline is the first of its kind in the Pakistani carpooling space and requires deep integration with Google ML Kit and local regulatory knowledge of CNIC data structures. The Dual-Device GPS Proximity Validation and Map Snapping anti-fraud system addresses mileage manipulation proactively rather than reactively, a capability absent in all identified competitors. The integration of Easypaisa and JazzCash escrow as the payment backbone is locally anchored and cannot be replicated by international platforms without significant regulatory and commercial groundwork. Finally, the NLP-powered behavioral trust-score system, which continuously updates each user's reputation through sentiment analysis of ride reviews, creates a self-reinforcing safety ecosystem that becomes more robust with every completed ride.

5.4 Customer Segments

- Daily commuters in Karachi travelling fixed routes to offices, universities, or commercial areas.
- Vehicle owners (drivers) seeking to recover rising fuel and maintenance costs without commercial profit.
- Students and young professionals priced out of Careem or Uber for daily commuting expenses.
- Environmentally conscious commuters motivated by tangible carbon-credit rewards.
- General public (non-students, non-employees) excluded from platforms requiring institutional emails.
- Corporate employees seeking cost-sharing with colleagues travelling along the same route.

5.6 Channels

SmartPool will reach its users primarily through the Google Play Store as a publicly downloadable mobile application, making it accessible to all Android users across Karachi without any registration barriers. Digital acquisition will leverage social media advertising on Instagram, Facebook, and TikTok, targeting commuter demographics and university communities. WhatsApp broadcast groups, university noticeboards, and corporate employee groups will serve as organic grassroots channels. A referral programme that rewards successful invitations with bonus carbon credits will accelerate peer-to-peer word-of-mouth growth. Partnerships with university transport offices and corporate HR departments will provide direct access to structured commuter communities with known, fixed routes ideal for carpooling matching.

5.7 Cost Structure

The primary cost drivers for SmartPool are technical human resources and cloud infrastructure. The four-member development team working across the Flutter/React Native frontend, Node.js backend, AI/ML optimization engine, and security pipeline constitutes the single largest cost at Rs. 1,080,000 over the nine-month development cycle. Cloud and API services including VPS hosting, Google Maps Platform quota, Firebase, SMS OTP gateway, and SSL certification account for Rs. 345,000. Equipment costs for the development laptop and Android test device total Rs. 200,000. Support staff including the academic supervisor, QA coordinator, and documentation writer add Rs. 180,000. Google Play Store publishing, App Store Optimization, and initial launch marketing contribute Rs. 45,000. A contingency fund of Rs. 100,000 covers unforeseen API overages or hardware failures. The total project budget amounts to Rs. 2,175,000.

5.8 Revenue Streams

SmartPool's primary revenue mechanism is a fixed platform base fee (BF = Rs. 30) embedded within every trip's total cost calculation, ensuring revenue is generated on each completed ride without charging passengers separately. This fee covers secure user verification, cloud hosting, and escrow processing costs. A Premium Driver Subscription at Rs. 299 per month provides high-frequency drivers with priority route matching, advanced trip analytics, and enhanced profile visibility. A Carbon Credit Marketplace allows users to redeem their earned green credits for partner fuel discounts, retail vouchers, and utility rebates, with SmartPool earning a commission on redemptions. Corporate Carpooling Packages offer organisations bulk-registered accounts with centralised billing, generating B2B revenue from companies wishing to formalise employee ride-sharing. Future revenue will be derived from anonymised, aggregated commute analytics sold to city planners, transit authorities, and urban research institutions to inform infrastructure decisions.

5.9 Business Model Canvas

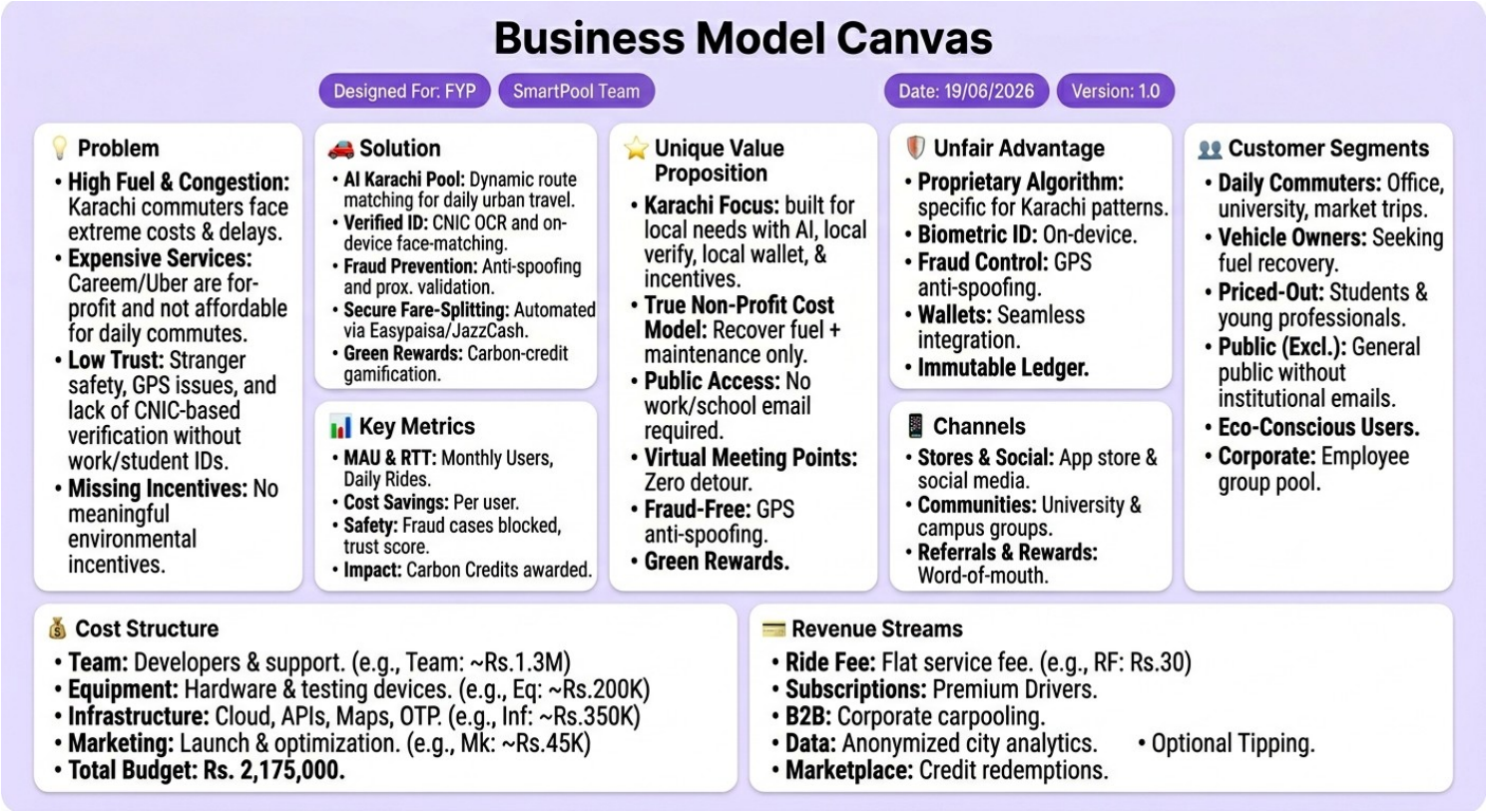


Fig 1. Business Model Canvas for SmartPool — Smart Carpooling System

Chapter #6

POSTER DESIGN



Brochure Design

